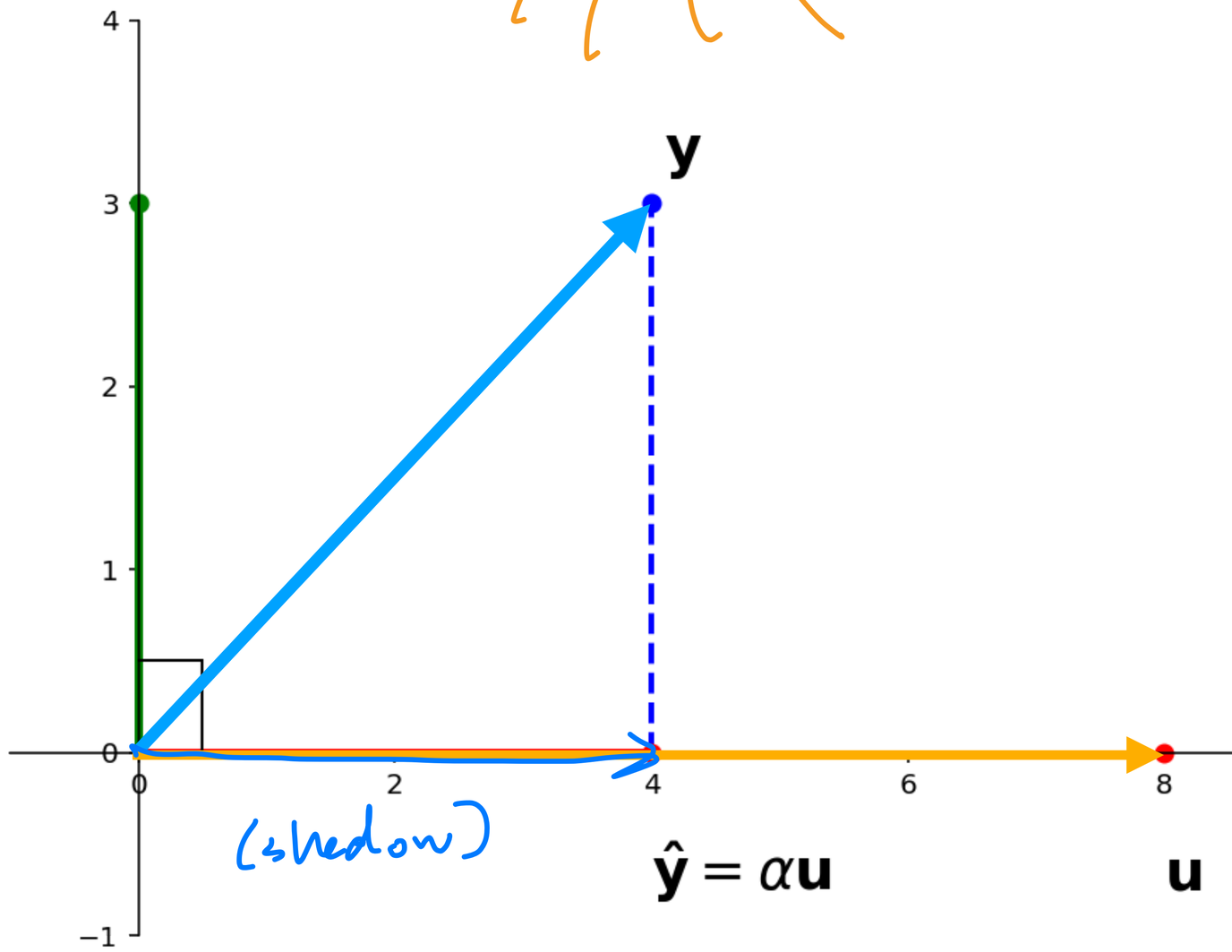
Least Squares

Geometric Algorithms

CAS CS 132

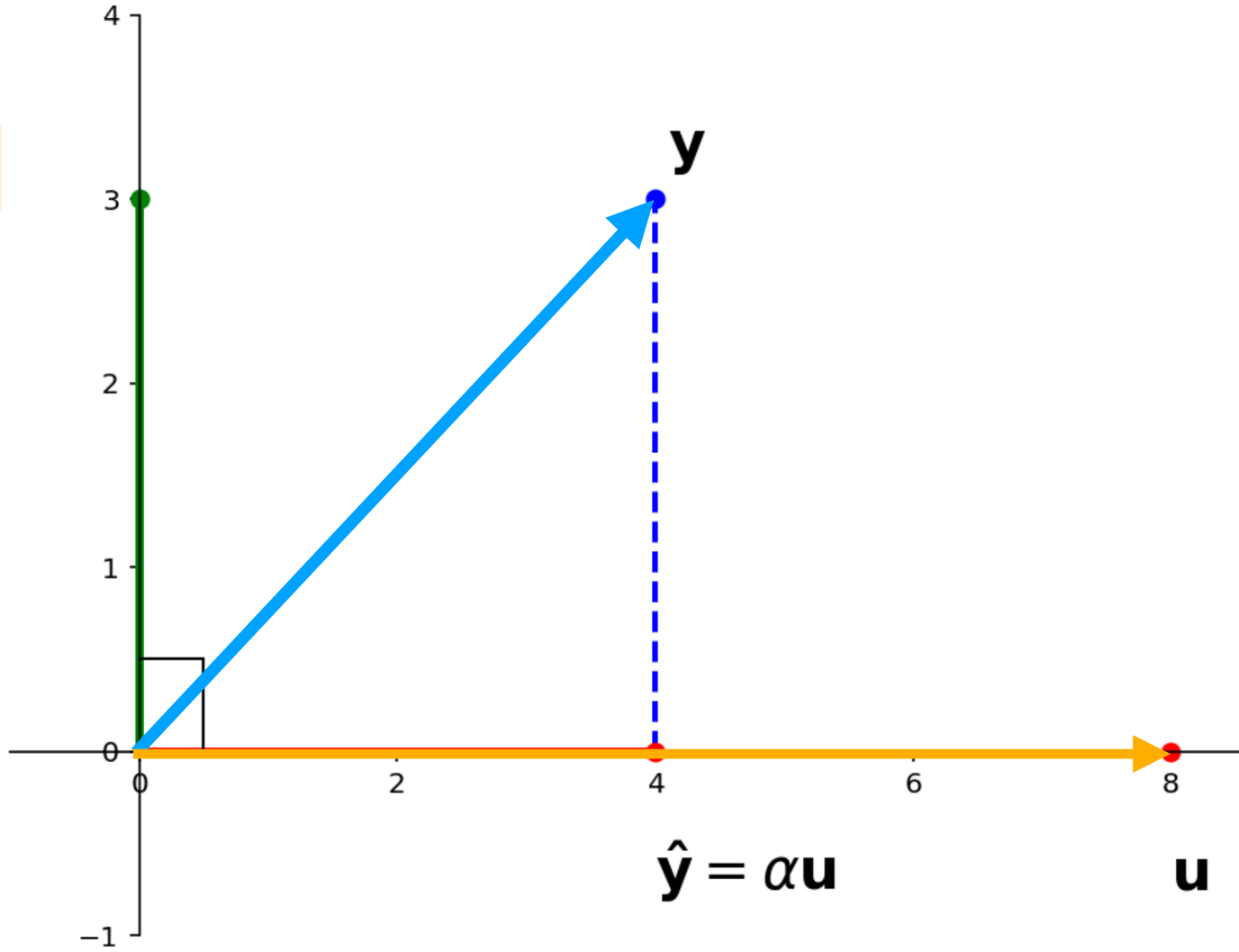
Recap

Recall: Orthogonal Projection



Recall: Orthogonal Projection

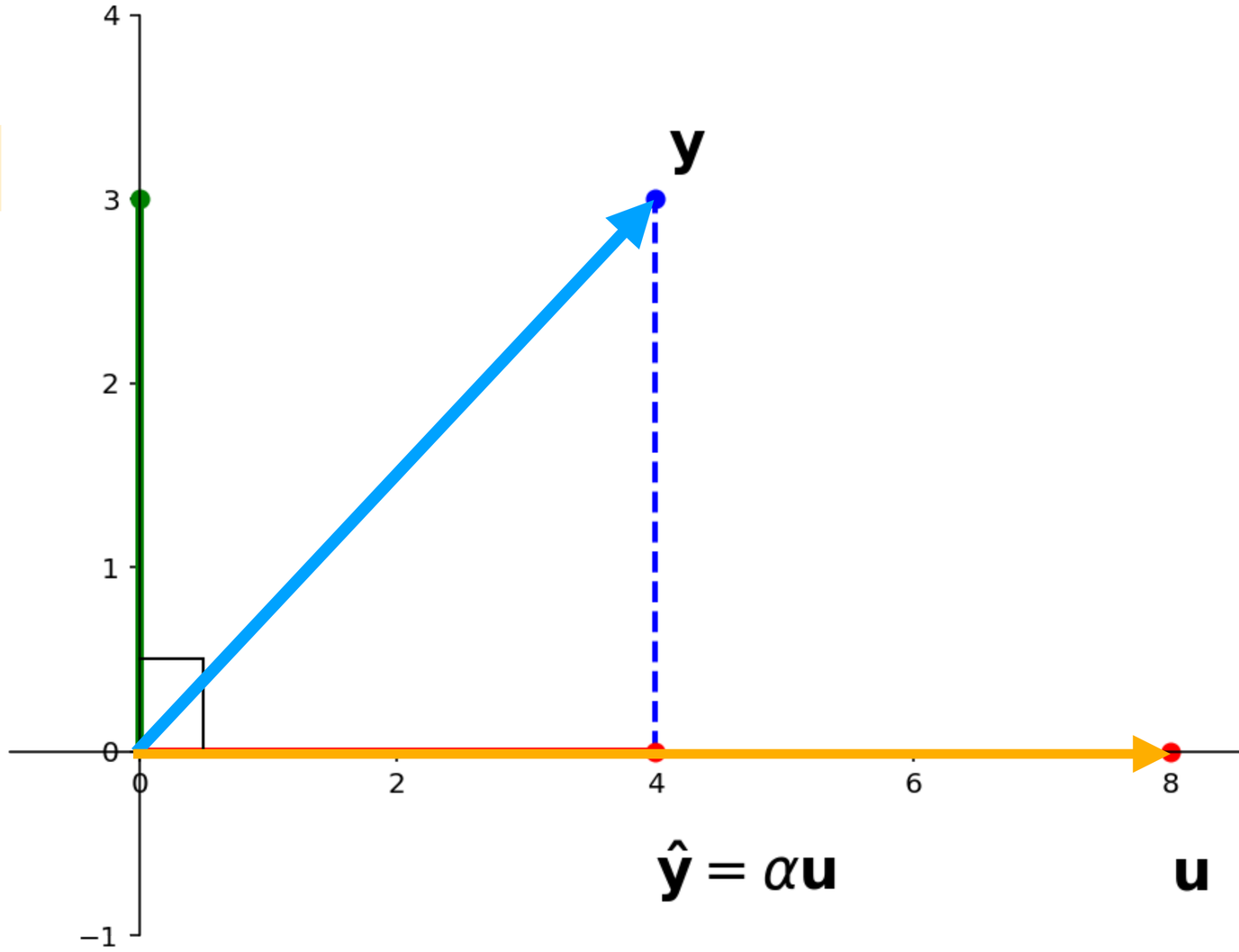
Given vectors $\mathbf{y}$ and $\mathbf{u}$ in $\mathbb{R}^n$, find vectors $\hat{\mathbf{y}}$ and $\mathbf{z}$ such that:



Recall: Orthogonal Projection

Given vectors $\mathbf{y}$ and $\mathbf{u}$ in $\mathbb{R}^n$, find vectors $\hat{\mathbf{y}}$ and $\mathbf{z}$ such that:

» $\mathbf{z}$ is orthogonal to $\mathbf{u}$ (i.e., $\mathbf{z} \cdot \mathbf{u} = 0$)

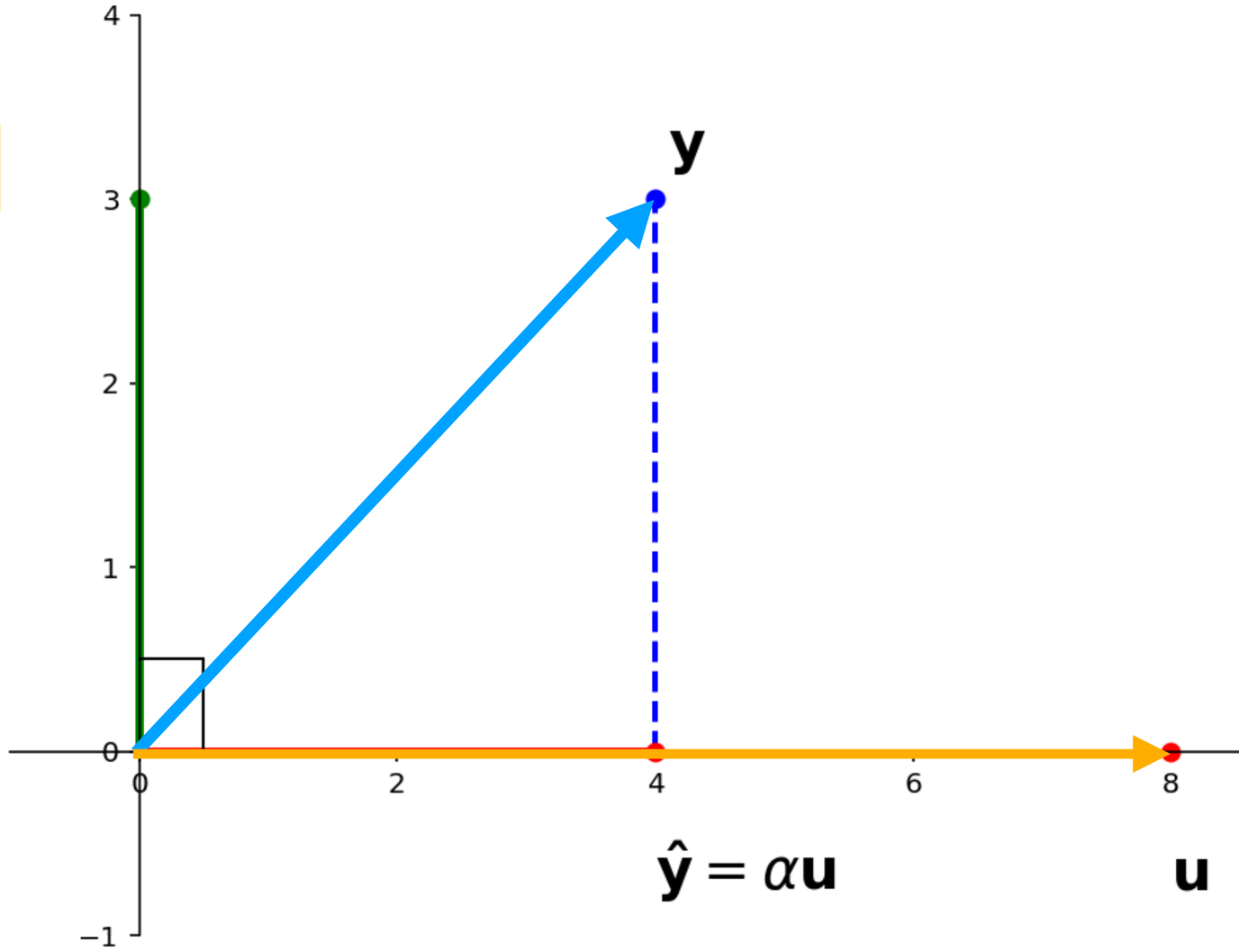


Recall: Orthogonal Projection

Given vectors $\mathbf{y}$ and $\mathbf{u}$ in $\mathbb{R}^n$, find vectors $\hat{\mathbf{y}}$ and $\mathbf{z}$ such that:

» $\mathbf{z}$ is orthogonal to $\mathbf{u}$ (i.e., $\mathbf{z} \cdot \mathbf{u} = 0$)

» $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$



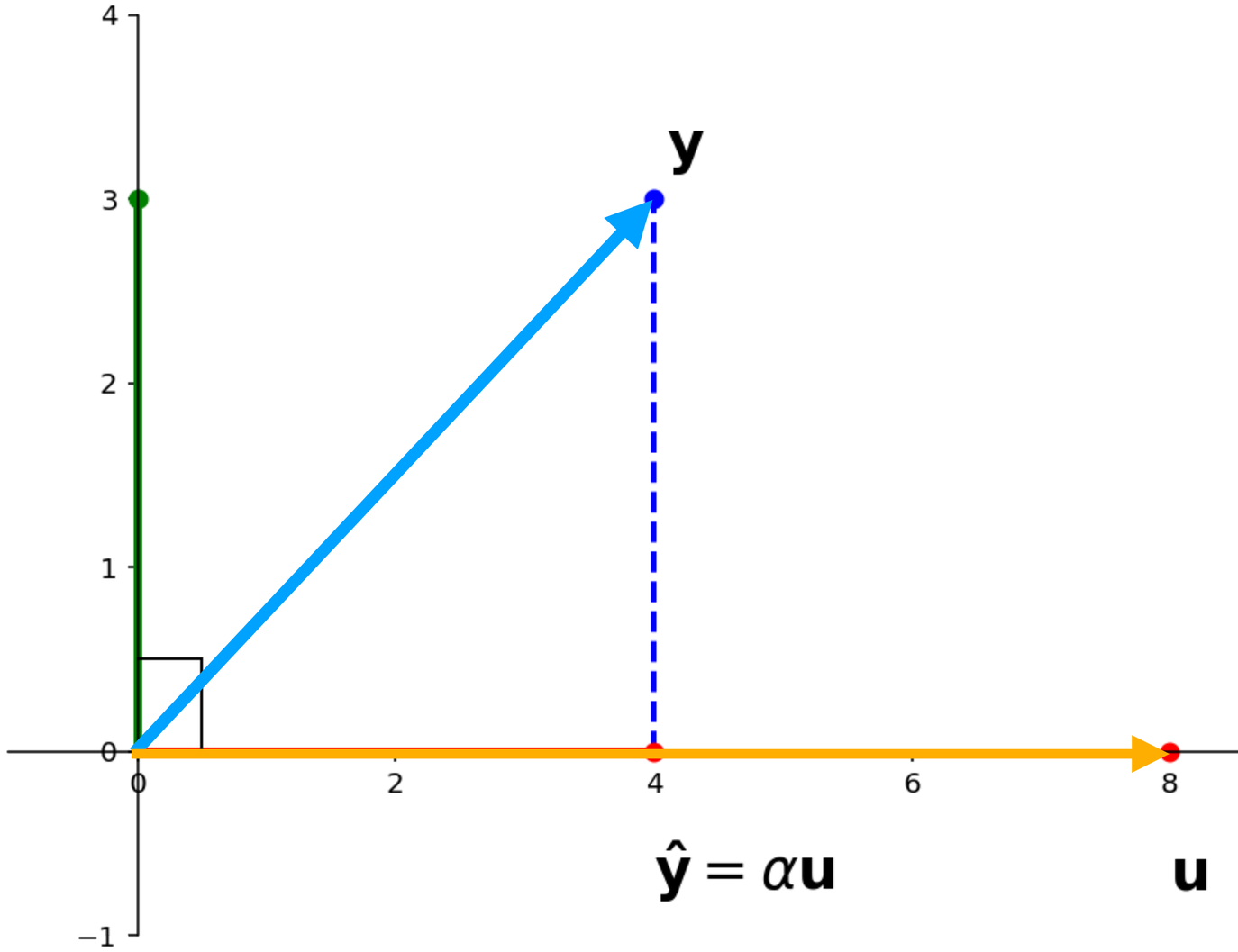
Recall: Orthogonal Projection

Given vectors $\mathbf{y}$ and $\mathbf{u}$ in $\mathbb{R}^n$, find vectors $\hat{\mathbf{y}}$ and $\mathbf{z}$ such that:

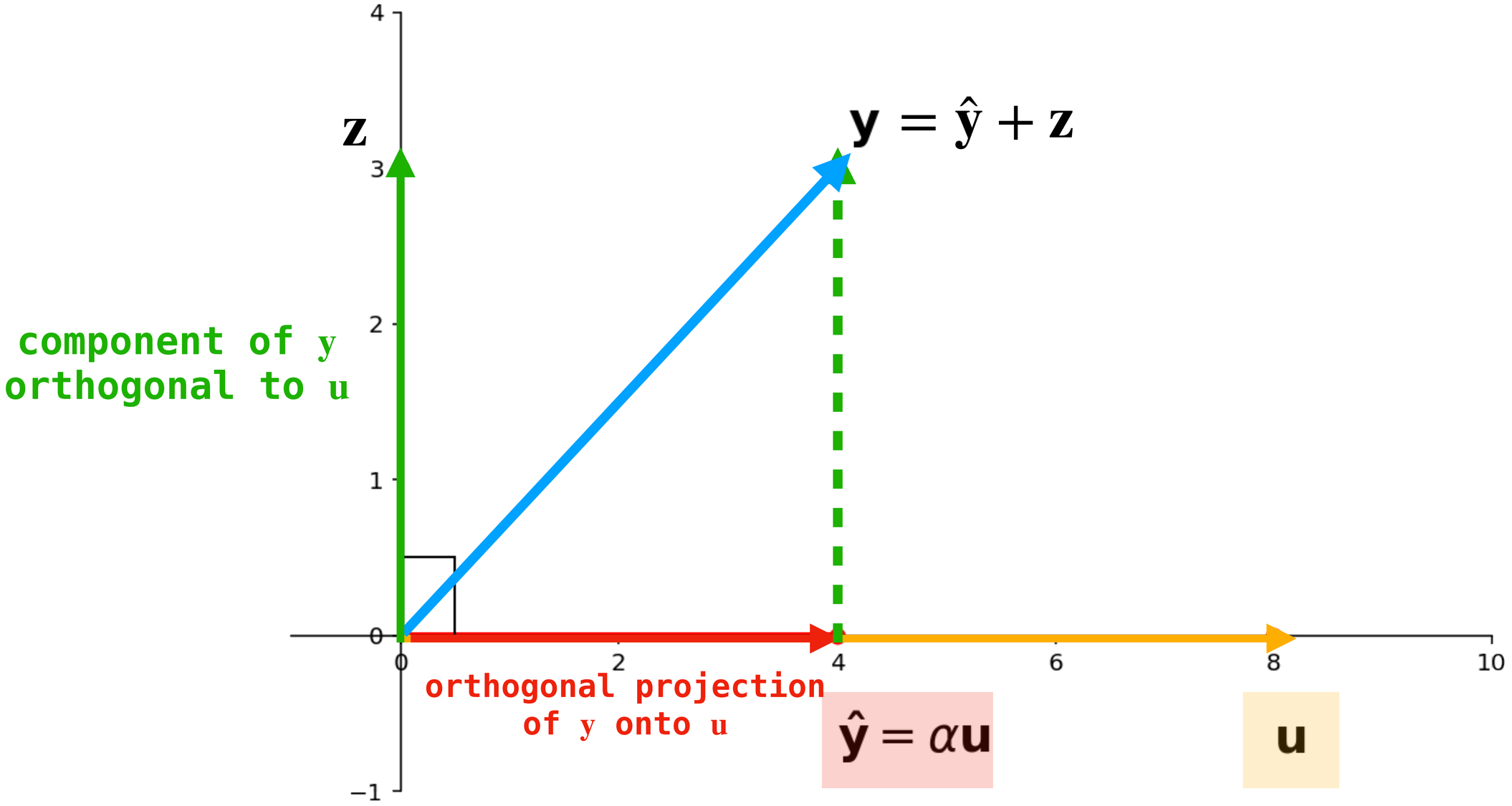
» $\mathbf{z}$ is orthogonal to $\mathbf{u}$ (i.e., $\mathbf{z} \cdot \mathbf{u} = 0$)

» $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$

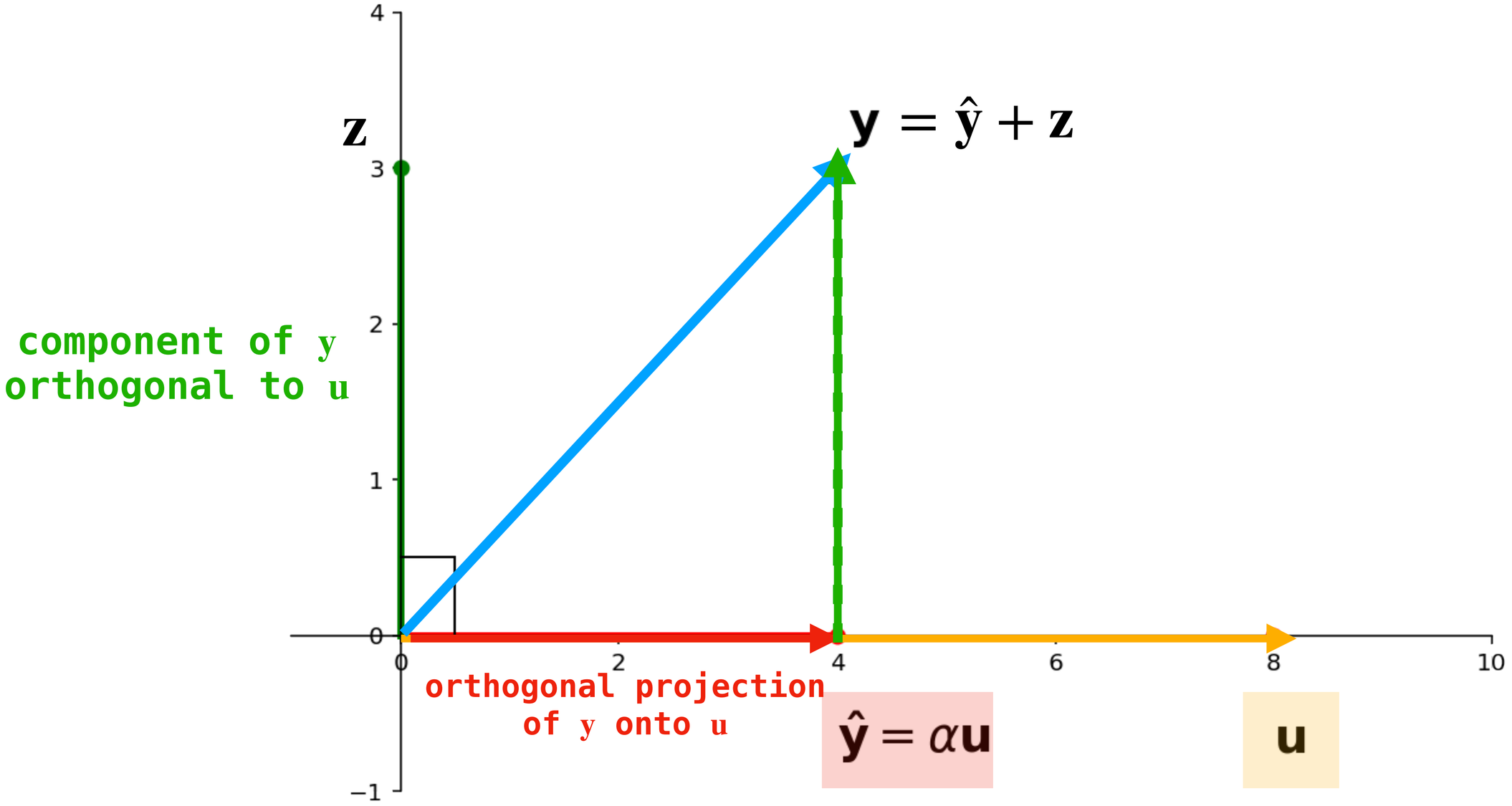
» $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$



Orthogonal Projection

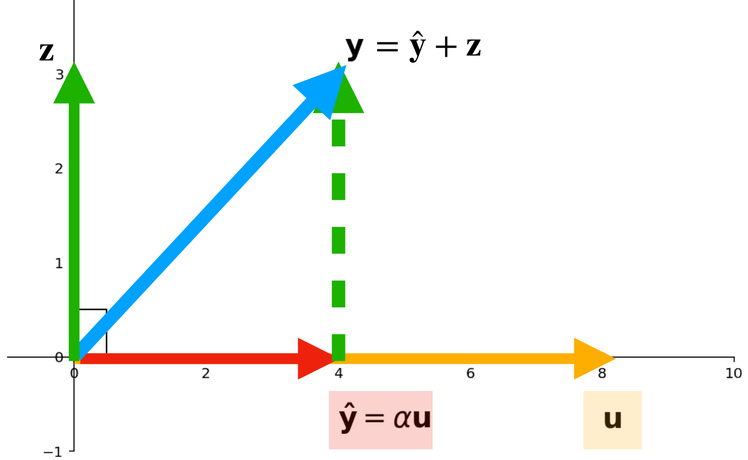


Orthogonal Projection

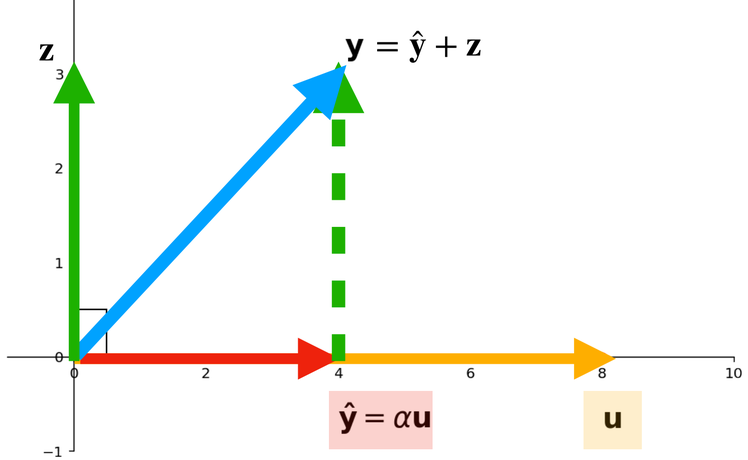


How do we find the orthogonal projection and orthogonal component?

What we know

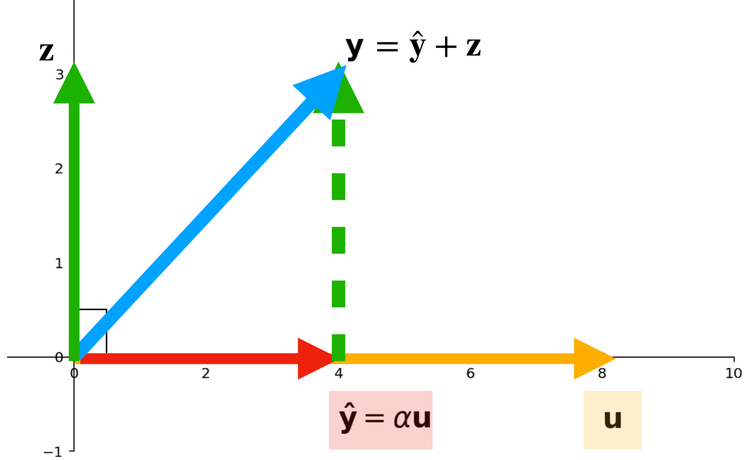


What we know



» $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$)

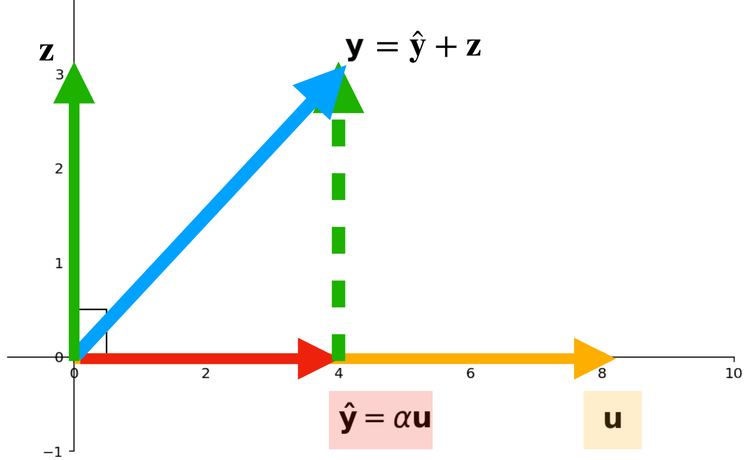
What we know



» $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$)

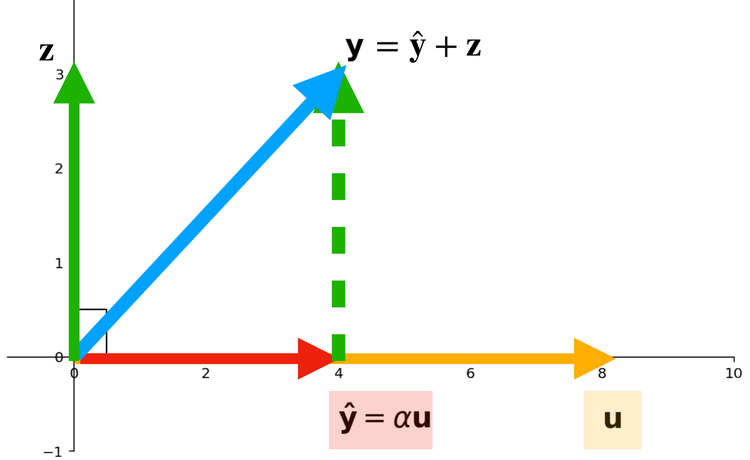
» $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \alpha\mathbf{u}$ (since $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$)

What we know



- $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$)
- $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \alpha\mathbf{u}$ (since $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$)
- $\langle \mathbf{z}, \mathbf{u} \rangle = 0$ (since $\mathbf{z}$ is orthogonal with $\mathbf{u}$)

What we know



» $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$)

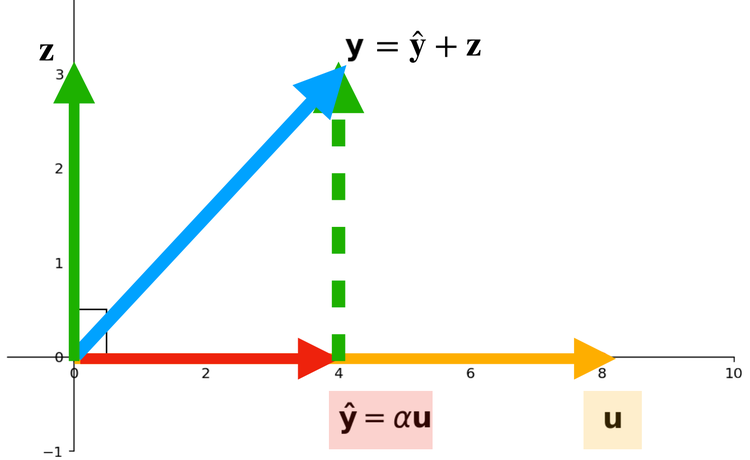
» $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \alpha\mathbf{u}$ (since $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$)

» $\langle \mathbf{z}, \mathbf{u} \rangle = 0$ (since $\mathbf{z}$ is orthogonal with $\mathbf{u}$)

Therefore:

$$\langle \mathbf{y} - \alpha\mathbf{u}, \mathbf{u} \rangle = 0$$

What we know



» $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in span\{\mathbf{u}\}$)

» $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \alpha\mathbf{u}$ (since $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$)

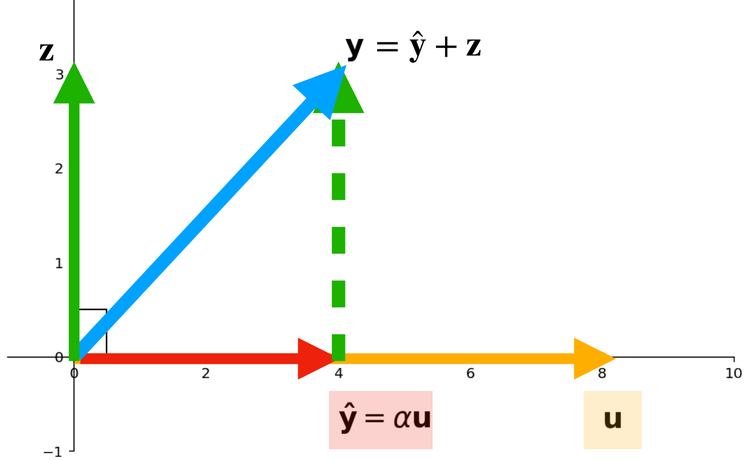
» $\langle \mathbf{z}, \mathbf{u} \rangle = 0$ (since $\mathbf{z}$ is orthogonal with $\mathbf{u}$)

Therefore:

$$\langle \mathbf{y} - \alpha\mathbf{u}, \mathbf{u} \rangle = 0$$

Once we have α, we can compute both $\hat{\mathbf{y}}$ and $\mathbf{z}$

Finding α



$$\langle \mathbf{y} - \alpha\mathbf{u}, \mathbf{u} \rangle = 0$$

$$\langle y - \alpha u, u \rangle = \langle y, u \rangle - \langle \alpha u, u \rangle$$

$$= \langle y, u \rangle - \alpha \langle u, u \rangle$$

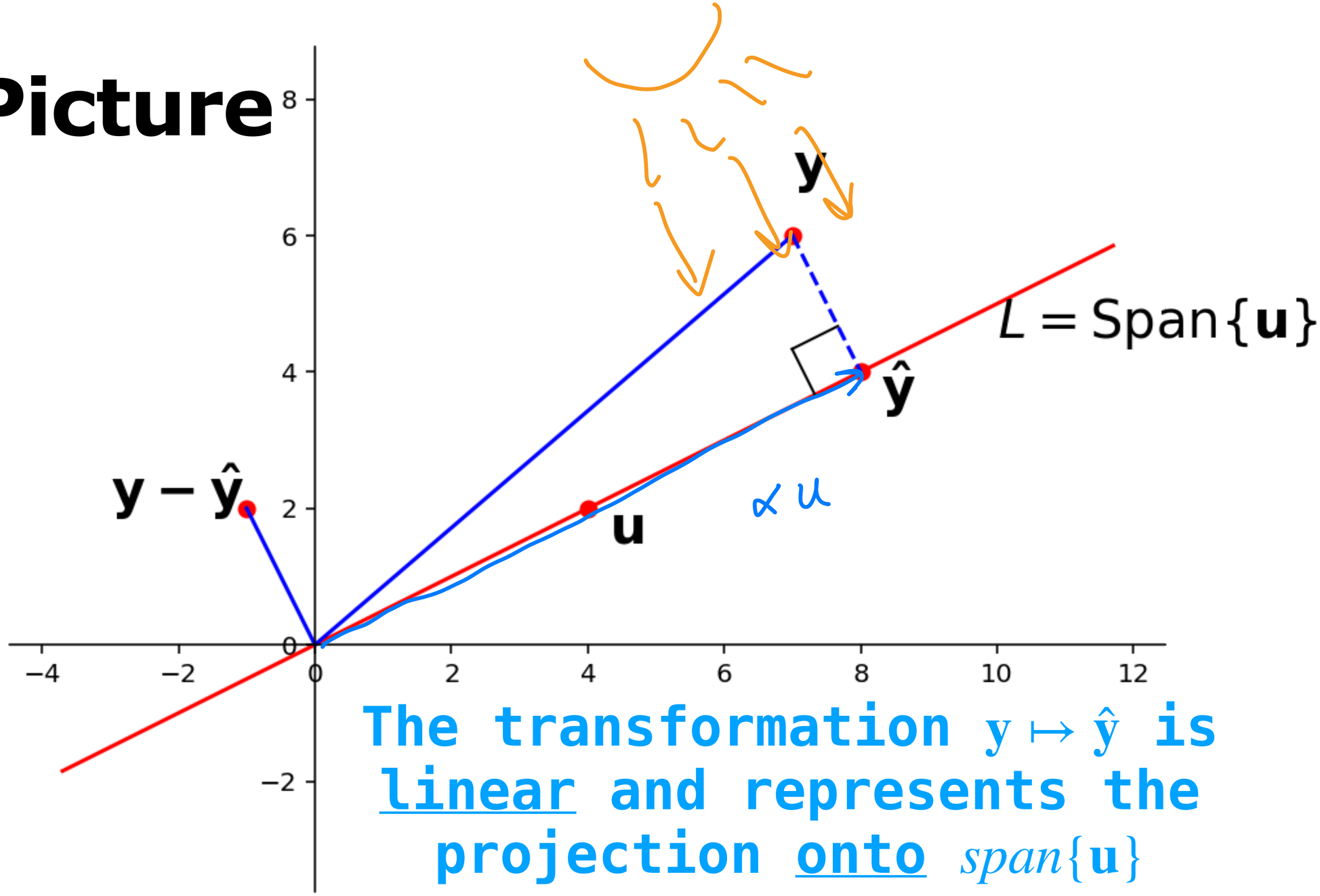
$$= 0$$

$$\alpha = \frac{\langle y, u \rangle}{\langle u, u \rangle}$$

$$\hat{y} = \frac{\langle y, u \rangle u}{\langle u, u \rangle}$$

$$\vec{z} = y - \hat{y} = \dots$$

The Picture



The transformation $\mathbf{y} \mapsto \hat{\mathbf{y}}$ is linear and represents the projection onto $span\{\mathbf{u}\}$

Recap Problem

$$\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

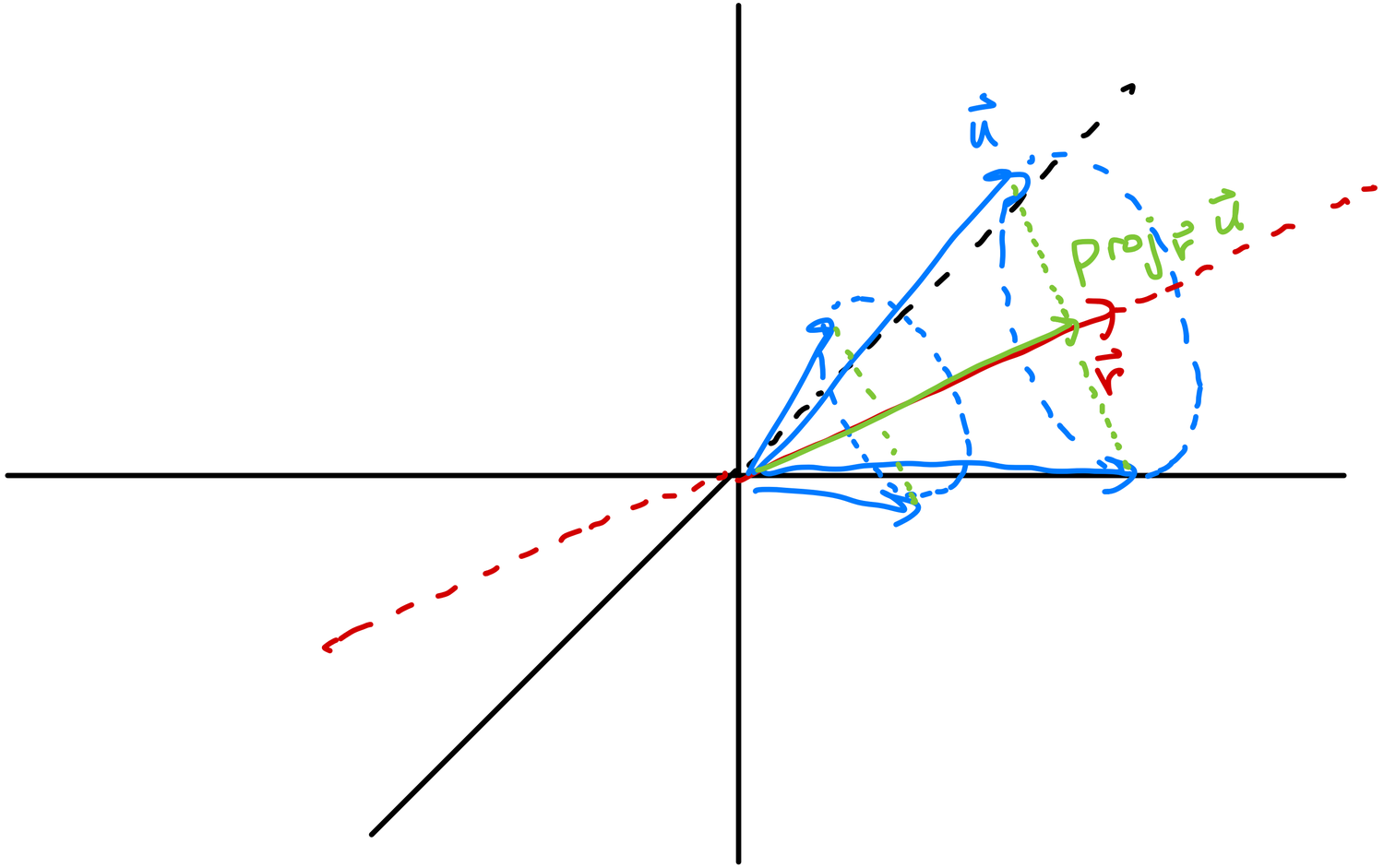
Find the orthogonal projection of $\mathbf{u}$ *onto the span of* $\mathbf{v}$

Find the matrix which implements orthogonal projection onto the span of $\mathbf{v}$

Answer

$$\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

`



Answer

$$\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$\langle u, v \rangle = 1(0) + 3(1) + (-2)(-1) + (-1)(0)$

$= 0 + 3 + 2 + 0 = 5$

$\langle v, v \rangle = 0^2 + 1^2 + (-1)^2 + 0^2 = 2$

$$\hat{u} = \alpha \vec{v} = \frac{\langle u, v \rangle}{\langle v, v \rangle} \vec{v} = \begin{bmatrix} 0 \\ 5/2 \\ -5/2 \\ 0 \end{bmatrix}$$

Answer

$$\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$\langle v, v \rangle = 2$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \mapsto \frac{\langle e_1, v \rangle}{\langle v, v \rangle} \vec{v} = \frac{0}{2} \vec{v} = \vec{0}$$

$$\vec{e}_2 \mapsto \frac{\langle e_2, v \rangle}{\langle v, v \rangle} \vec{v} = \frac{1}{2} \vec{v} = \begin{bmatrix} 0 \\ 1/2 \\ -1\,1/2 \\ 0 \end{bmatrix}$$

$$\vec{e}_3 \mapsto \frac{\langle e_3, v \rangle}{\langle v, v \rangle} \vec{v} = \frac{-1}{2} \vec{v} = \begin{bmatrix} 0 \\ -1/2 \\ 1/2 \\ 0 \end{bmatrix}$$

$$\vec{e}_4 \mapsto \vec{0}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1/2 & -1/2 & 0 \\ 0 & -1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 3/2 + 2/2 \\ -3/2 - 2/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5/2 \\ -5/2 \\ 0 \end{bmatrix}$$

Objectives

Introduce the **least squares problem** as a method of *approximating* solutions to matrix equations

Learn how to solve the least squares problems

Connect least squares solutions to projections

Keywords

general least squares problem

sum of squares error (ℓ_2–error)

least squares solutions

orthogonal projections

normal equations

Motivation

The story of an enterprising CS132 student

The story of an enterprising CS132 student

Q. Solve the equation $A\mathbf{x} = \mathbf{b}$

The story of an enterprising CS132 student

Q. Solve the equation $A\mathbf{x} = \mathbf{b}$

A. Use `np.linalg.solve(A, b)`

The story of an enterprising CS132 student

Q. Solve the equation $A\mathbf{x} = \mathbf{b}$

A. Use np.linalg.solve(A, b)

```
>>> A = np.array([
...     [1., 0, 5],
...     [1, -1, 4],
...     [0, 2, 2]])
>>> b = np.array([-1, 2, 3])
>>> np.linalg.solve(A, b)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/opt/homebrew/lib/python3.11/site-packages/numpy/linalg/linalg.py", line 409, in solve
    r = gufunc(a, b, signature=signature, extobj=extobj)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/opt/homebrew/lib/python3.11/site-packages/numpy/linalg/linalg.py", line 112, in _raise_linalgerror_singular
    raise LinAlgError("Singular matrix")
numpy.linalg.LinAlgError: Singular matrix
```

The story of an enterprising CS132 student

Q. Solve the equation $A\mathbf{x} = \mathbf{b}$

A. Use np.linalg.solve(A, b)

```
>>> A = np.array([
...     [1., 0, 5],
...     [1, -1, 4],
...     [0, 2, 2]])
>>> b = np.array([-1, 2, 3])
>>> np.linalg.solve(A, b)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/opt/homebrew/lib/python3.11/site-packages/numpy/linalg/linalg.py", line 409, in solve
    r = gufunc(a, b, signature=signature, extobj=extobj)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/opt/homebrew/lib/python3.11/site-packages/numpy/linalg/linalg.py", line 112, in _raise_linalgerror_singular
    raise LinAlgError("Singular matrix")
numpy.linalg.LinAlgError: Singular matrix
```

This doesn't always work

Reads the docs...

numpy.linalg.solve

linalg.solve(*a*, *b*) [source]

Solve a linear matrix equation, or system of linear scalar equations.

Computes the “exact” solution, *x*, of the well-determined, i.e., full rank, linear matrix equation *ax* = *b*.

Parameters: **a** : *(..., M, M) array_like*
Coefficient matrix.

b : *{(..., M,), (..., M, K)}, array_like*
Ordinate or “dependent variable” values.

Returns: **x** : *{(..., M,), (..., M, K)} ndarray*
Solution to the system a x = b. Returned shape is identical to *b*.

Raises: **LinAlgError**
If *a* is singular or not square.

See also

scipy.linalg.solve

Reads the docs...

numpy.linalg.solve

linalg.solve(*a*, *b*) [source]

Solve a linear matrix equation, or system of linear scalar equations.

Computes the “exact” solution, *x*, of the well-determined, i.e., full rank, linear matrix equation *ax* = *b*.

Parameters: **a** : *(..., M, M) array_like*
Coefficient matrix.

b : *{(..., M,), (..., M, K)}, array_like*
Ordinate or “dependent variable” values.

Returns: **x** : *{(..., M,), (..., M, K)} ndarray*
Solution to the system a x = b. Returned shape is identical to *b*.

Raises: **LinAlgError**
If *a* is singular or not square.

See also

scipy.linalg.solve

Reads the docs...

b : {(..., M,), (..., M, K)}, array_like

Ordinate or “dependent variable” values.

Returns: x : *{(..., M,), (..., M, K)} ndarray*

Solution to the system a x = b. Returned shape is identical to *b*.

Raises: LinAlgError

If *a* is singular or not square.

See also

`scipy.linalg.solve`

Similar function in SciPy.

Notes

New in version 1.8.0.

Broadcasting rules apply, see the `numpy.linalg` documentation for details.

The solutions are computed using LAPACK routine `_gesv`.

a must be square and of full-rank, i.e., all rows (or, equivalently, columns) must be linearly independent; if either is not true, use `lstsq` for the least-squares best “solution” of the system/equation.

Reads the docs...

b : {(..., M,), (..., M, K)}, array_like

Ordinate or “dependent variable” values.

Returns: x : *{(..., M,), (..., M, K)} ndarray*

Solution to the system a x = b. Returned shape is identical to *b*.

Raises: LinAlgError

If *a* is singular or not square.

See also

`scipy.linalg.solve`

Similar function in SciPy.

Notes

New in version 1.8.0.

Broadcasting rules apply, see the `numpy.linalg` documentation for details.

The solutions are computed using LAPACK routine `_gesv`.

a must be square and of full-rank, i.e., all rows (or, equivalently, columns) must be linearly independent; if either is not true, use `lstsq` for the least-squares best “solution” of the system/equation.

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

uh...probably numerical errors...

Final answer: $\mathbf{x} = \begin{bmatrix} -1/9 \\ 7/9 \\ 2/9 \end{bmatrix}$

np.linalg.lstsq

```
>>> np.linalg.lstsq(A, b)
<stdin>:1: FutureWarning: `rcond` parameter will change to the default of machine precision times ``max(M, N)``
where M and N are the input matrix dimensions.
To use the future default and silence this warning we advise to pass `rcond=None`, to keep using the old,
explicitly pass `rcond=-1`.
(array([-0.11111111,  0.77777778,  0.22222222]), array([], dtype=float64), 2, array([6.84168488e+00,
2.27845297e+00, 6.13801942e-17]))
>>> x = np.array([-0.11111111,  0.77777778,  0.22222222])
>>> A @ x
array([ 9.99999990e-01, -9.99999994e-09,  2.00000000e+00])
>>>
```

uh...probably numerical errors...

Final answer: $\mathbf{x} = \begin{bmatrix} -1/9 \\ 7/9 \\ 2/9 \end{bmatrix}$ **This is not correct**

This System is Inconsistent

$$\begin{bmatrix} 1 & 0 & 5 & -1 \\ 1 & -1 & 4 & 2 \\ 0 & 2 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & -1 \\ 0 & -1 & -1 & 3 \\ 0 & 2 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & -1 \\ 0 & -1 & -1 & 3 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

The correct answer: There is no solution

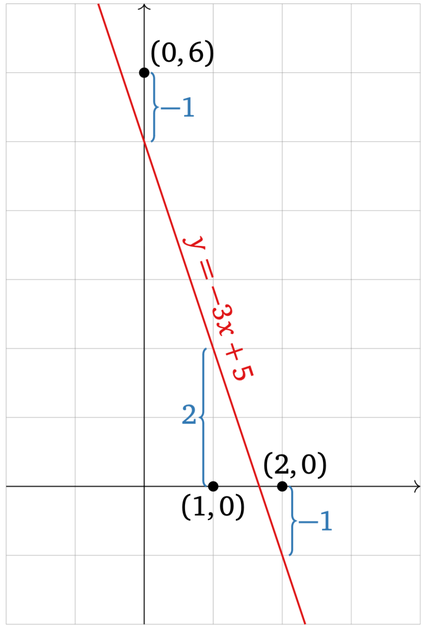
This System is Inconsistent

$$\begin{bmatrix} 1 & 0 & 5 & -1 \\ 1 & -1 & 4 & 2 \\ 0 & 2 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & -1 \\ 0 & -1 & -1 & 3 \\ 0 & 2 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & -1 \\ 0 & -1 & -1 & 3 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

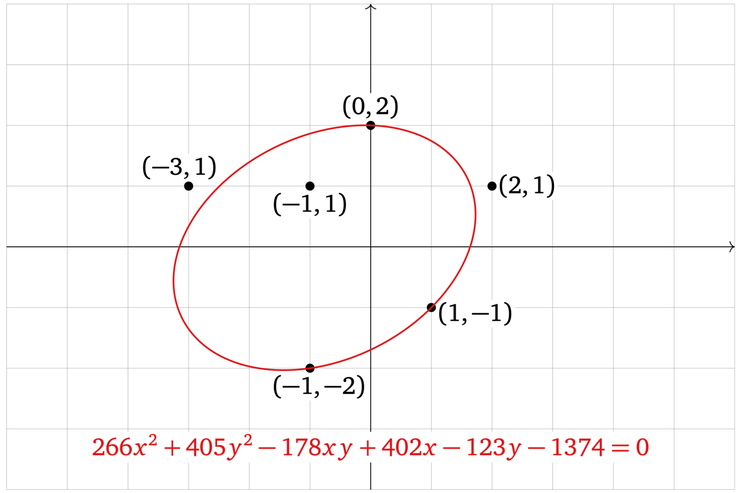
The correct answer: There is no solution

What's going on here?

Non-Linearity



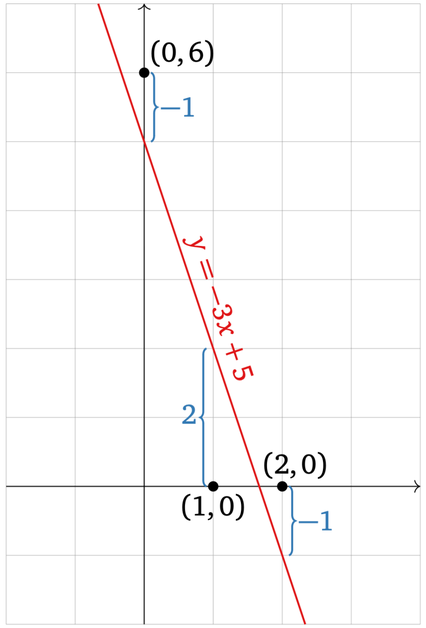
$$b - A\hat{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - A\begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$



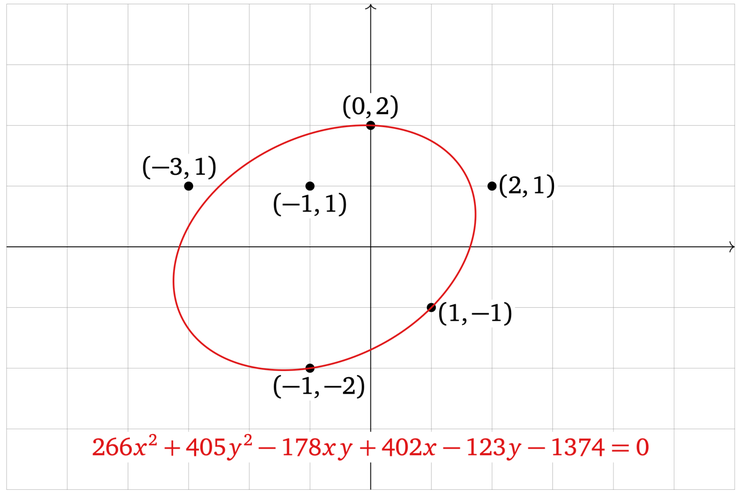
https://textbooks.math.gatech.edu/ila/least-squares.html

Non-Linearity

Linear algebra is powerful, but **the world isn't linear**



$$b - A\hat{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - A\begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

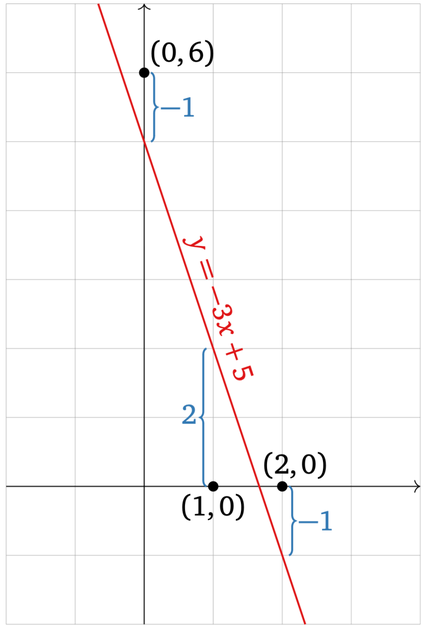


https://textbooks.math.gatech.edu/ila/least-squares.html

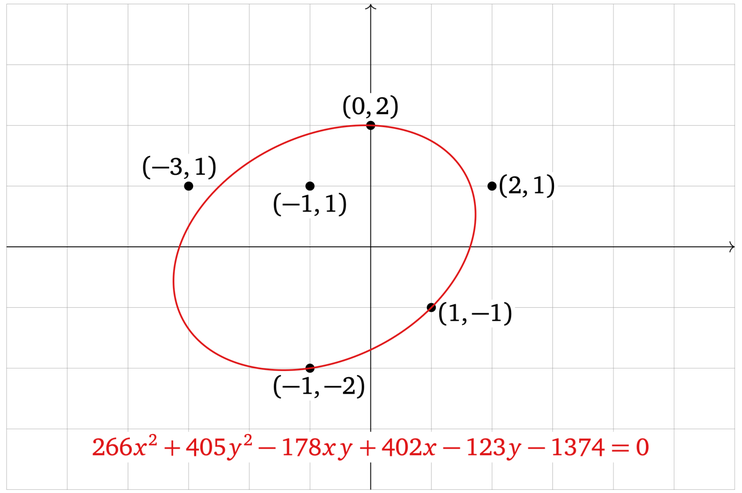
Non-Linearity

Linear algebra is powerful, but **the world isn't linear**

There's non-linear stuff



$$b - A\hat{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - A\begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$



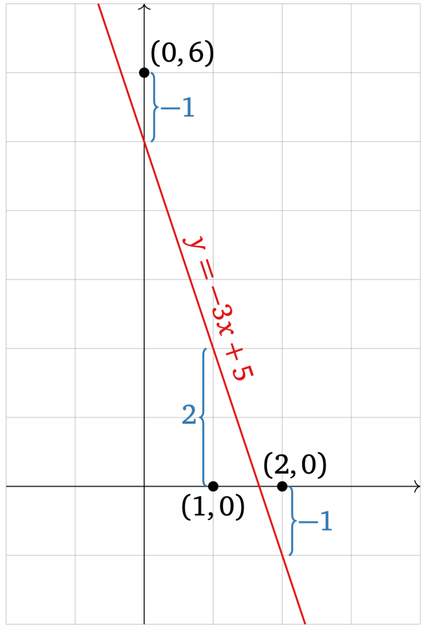
https://textbooks.math.gatech.edu/ila/least-squares.html

Non-Linearity

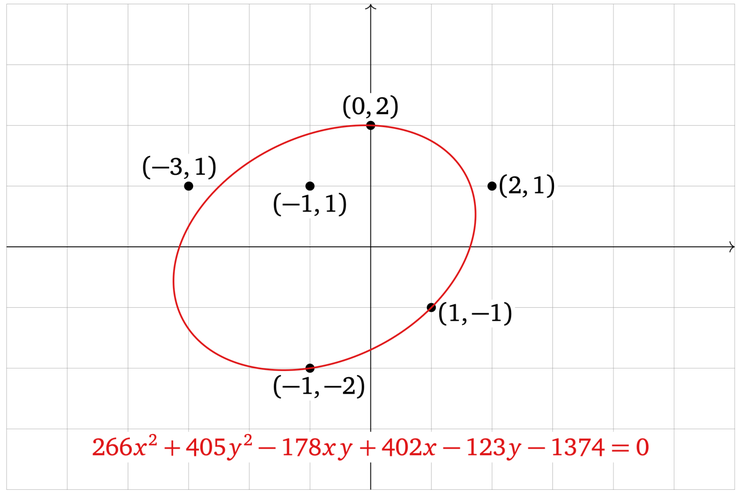
Linear algebra is powerful, but **the world isn't linear**

There's non-linear stuff

There's *noise*



$$b - A\hat{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - A\begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$



https://textbooks.math.gatech.edu/ila/least-squares.html

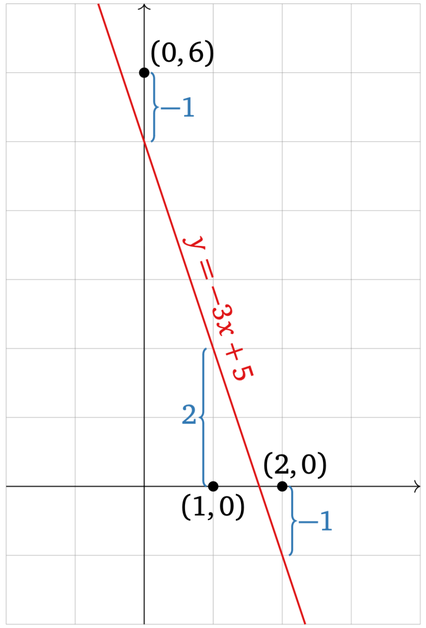
Non-Linearity

Linear algebra is powerful, but **the world isn't linear**

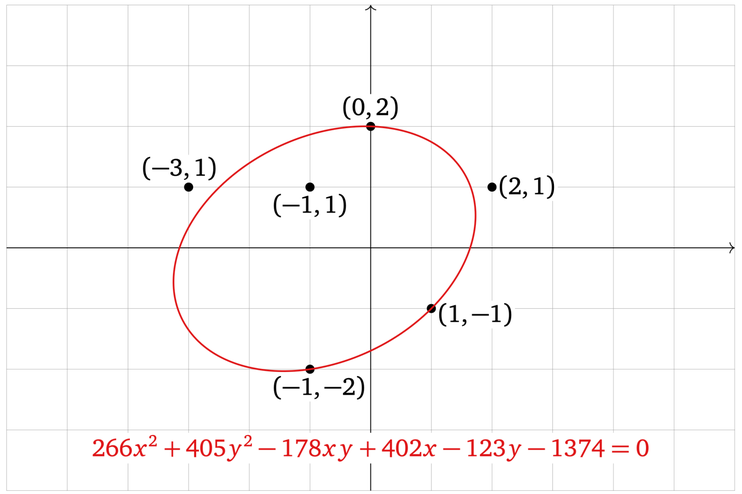
There's non-linear stuff

There's *noise*

We can't make the world linear



$$b - A\hat{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - A\begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$



https://textbooks.math.gatech.edu/ila/least-squares.html

Non-Linearity

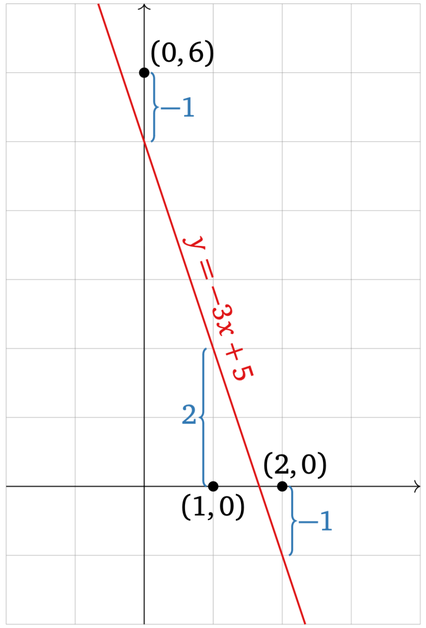
Linear algebra is powerful, but **the world isn't linear**

There's non-linear stuff

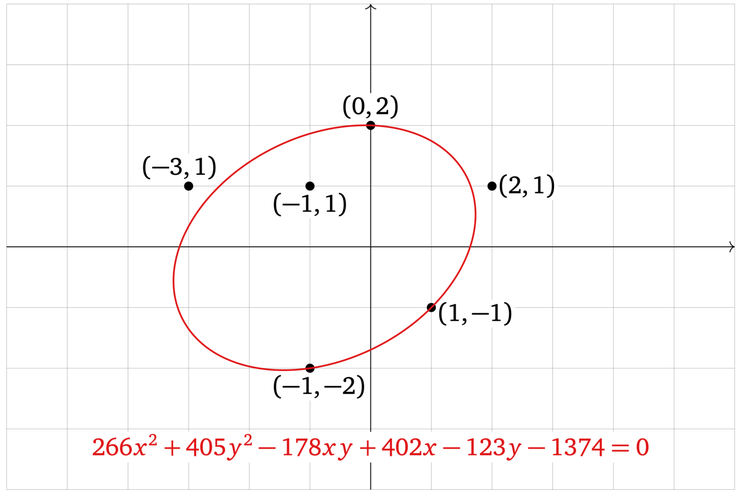
There's *noise*

We can't make the world linear

But we can pretend...

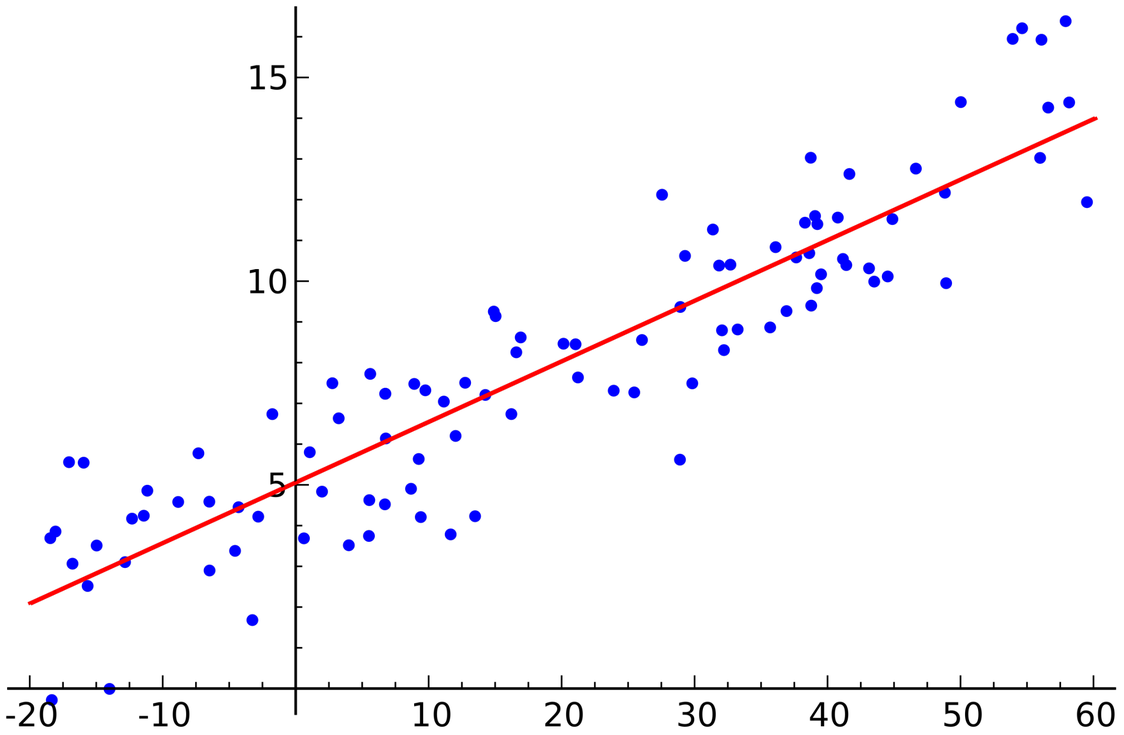


$$b - A\hat{x} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix} - A\begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$



https://textbooks.math.gatech.edu/ila/least-squares.html

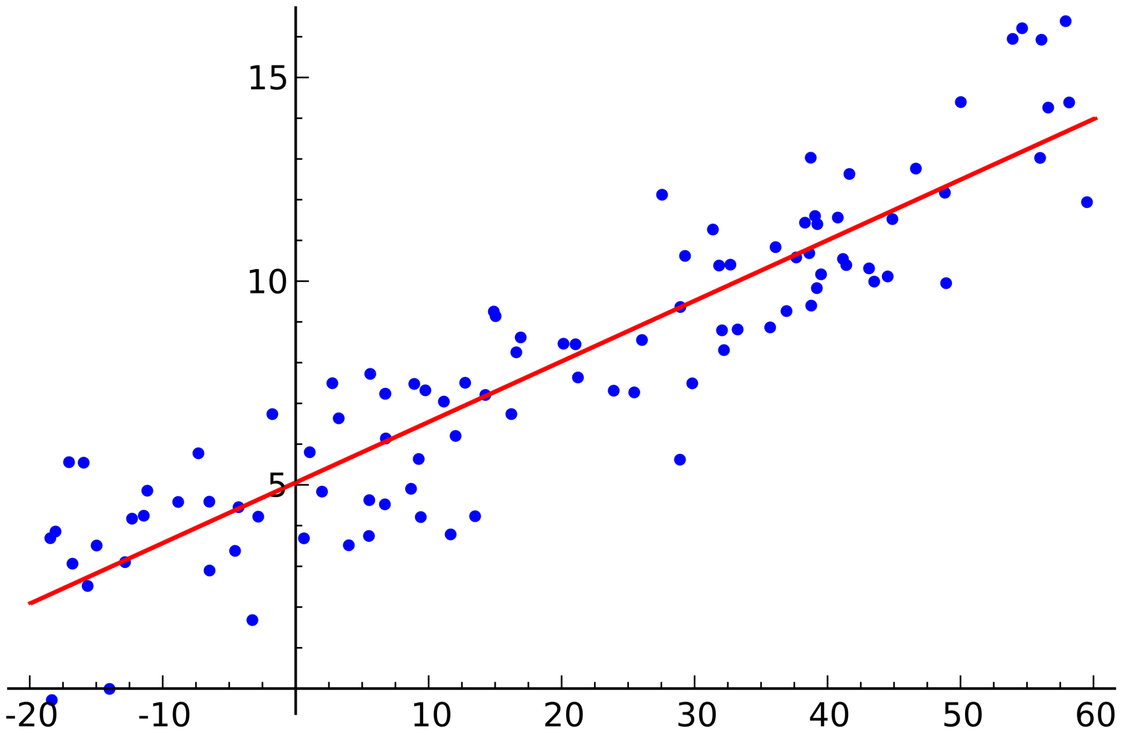
The Idea



https://commons.wikimedia.org/wiki/File:Linear_regression.svg

The Idea

Least Squares is a method for *approximating* solutions to systems of linear equations

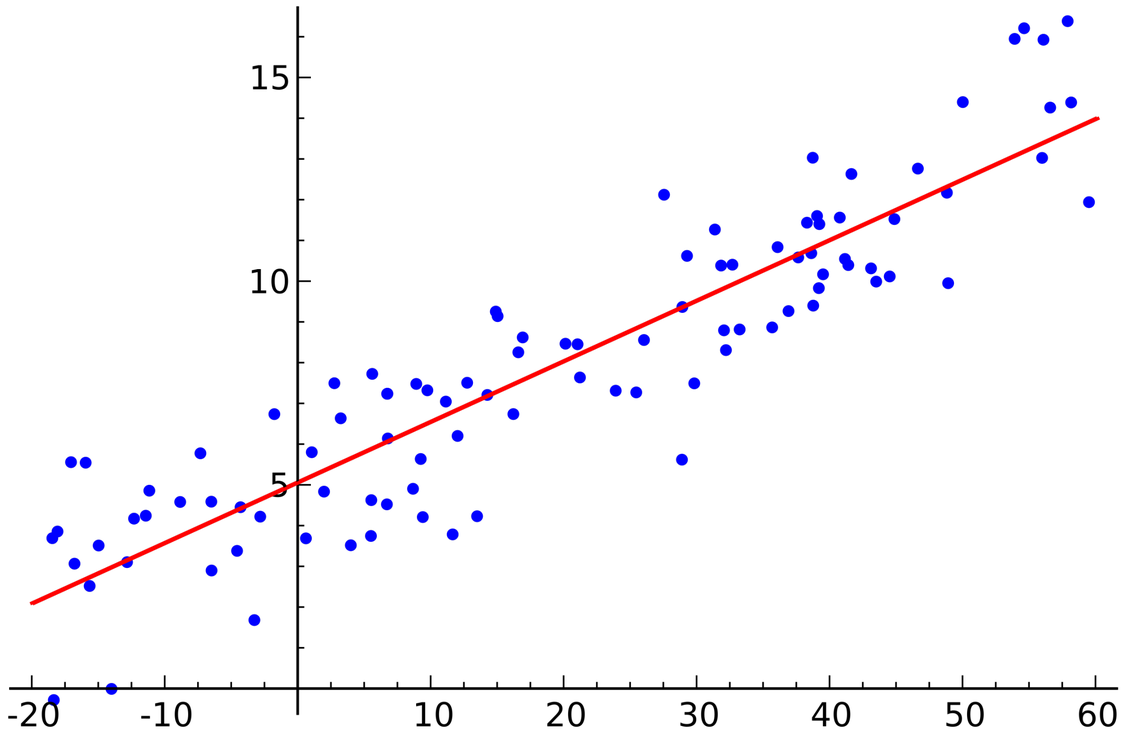


https://commons.wikimedia.org/wiki/File:Linear_regression.svg

The Idea

Least Squares is a method for *approximating* solutions to systems of linear equations

This is **more useful** than exact solutions



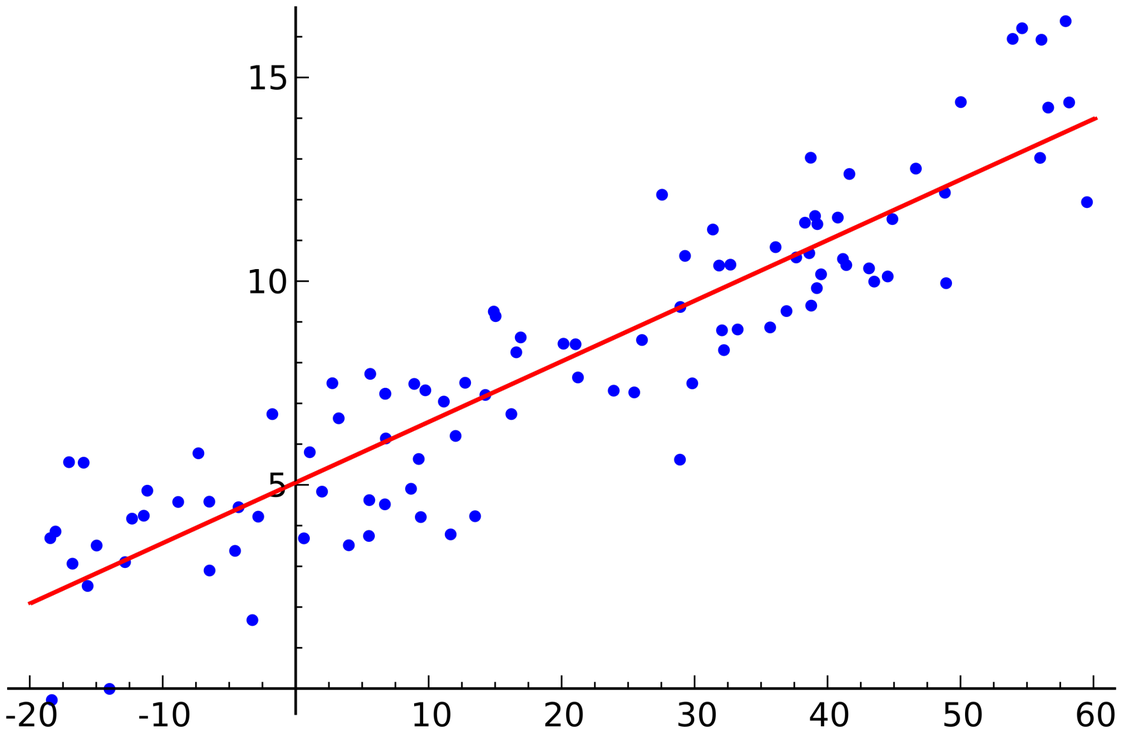
https://commons.wikimedia.org/wiki/File:Linear_regression.svg

The Idea

Least Squares is a method for *approximating* solutions to systems of linear equations

This is **more useful** than exact solutions

We can do **linear regression**



https://commons.wikimedia.org/wiki/File:Linear_regression.svg

General Least Squares Problem

The Picture

$A\hat{x} = b$

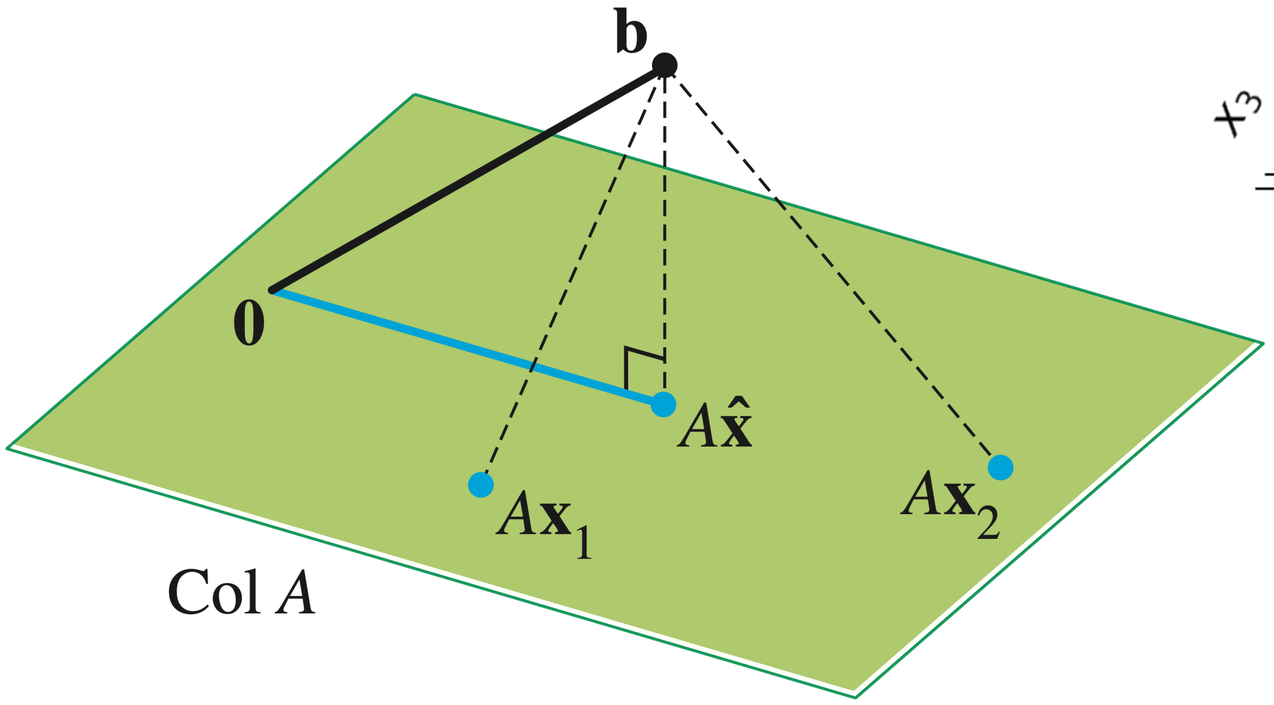
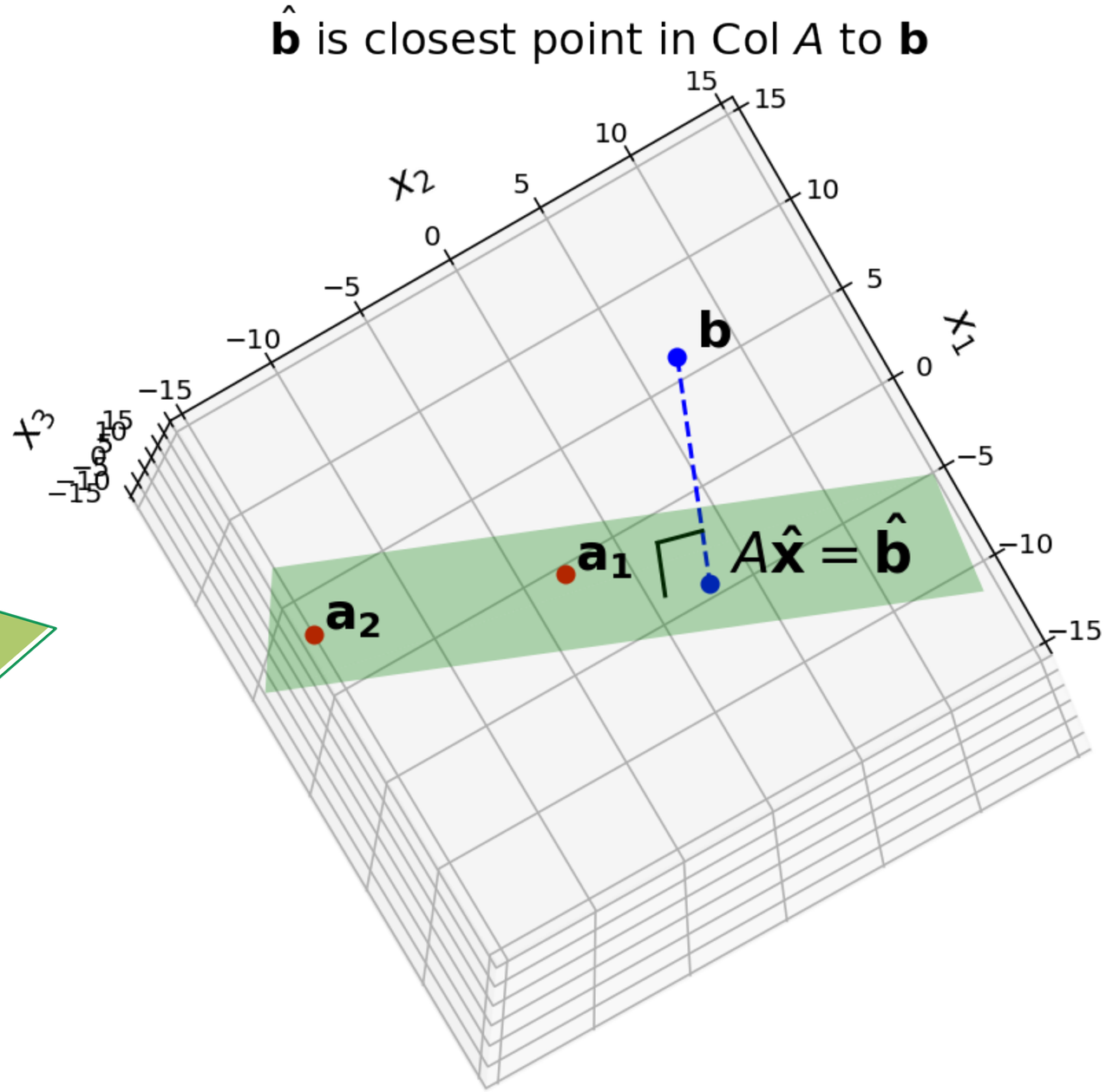


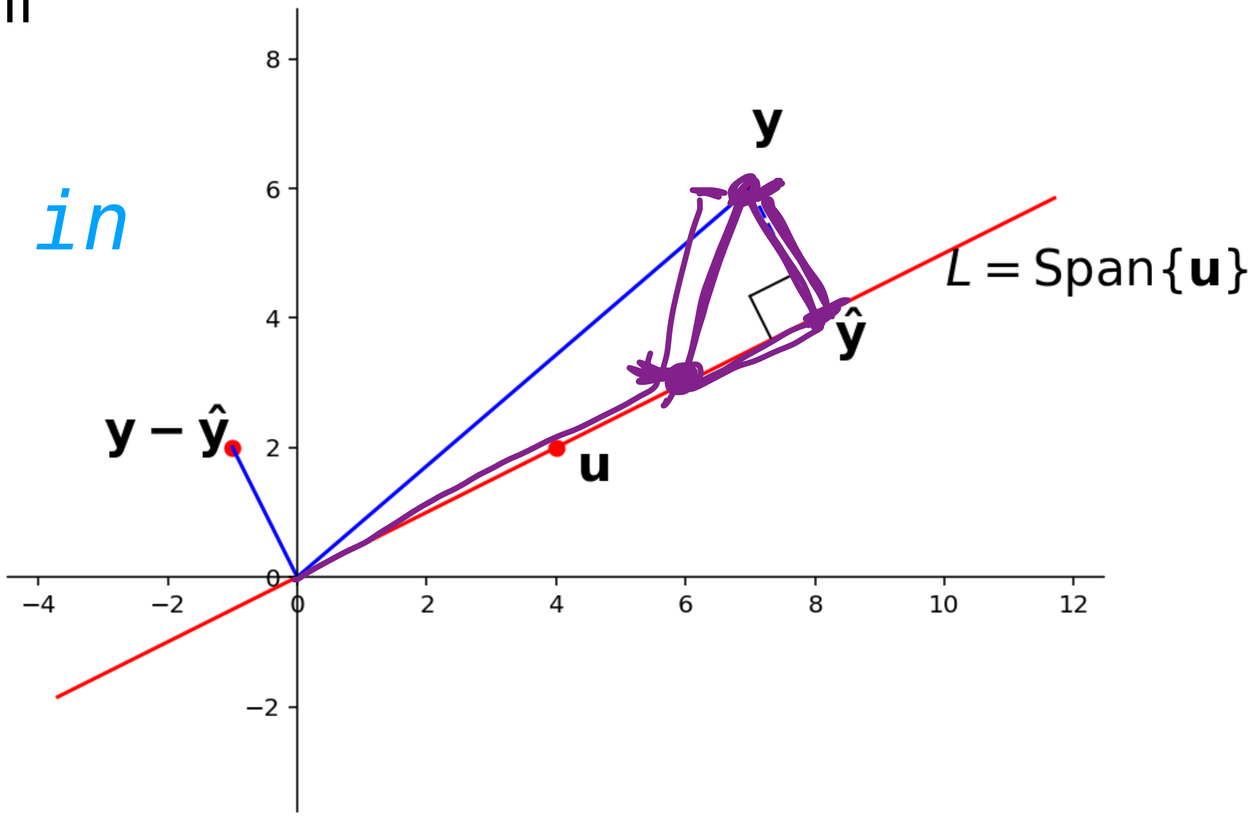
Figure 22.8



Recall: $\hat{\mathbf{y}}$ and Distance

Thm. $\|\hat{\mathbf{y}} - \mathbf{y}\| = \min_{\mathbf{w} \in span\{\mathbf{u}\}} \|\mathbf{w} - \mathbf{y}\|$

$\hat{\mathbf{y}}$ is the closest vector in $span\{\mathbf{u}\}$ to $\mathbf{y}$



The Equational Perspective

The Equational Perspective

We know the equation $x\mathbf{u} = \mathbf{y}$ may have no solution

The Equational Perspective

We know the equation $x\mathbf{u} = \mathbf{y}$ may have no solution

Q. Find α s.t. $\alpha\mathbf{u}$ is as close as possible to $\mathbf{y}$

The Equational Perspective

We know the equation $x\mathbf{u} = \mathbf{y}$ may have no solution

Q. Find α s.t. $\alpha\mathbf{u}$ is as close as possible to $\mathbf{y}$

i.e., $dist(\mathbf{y}, \alpha\mathbf{u}) = \|\mathbf{y} - \alpha\mathbf{u}\|$ is as small as possible

The Equational Perspective

We know the equation $x\mathbf{u} = \mathbf{y}$ may have no solution

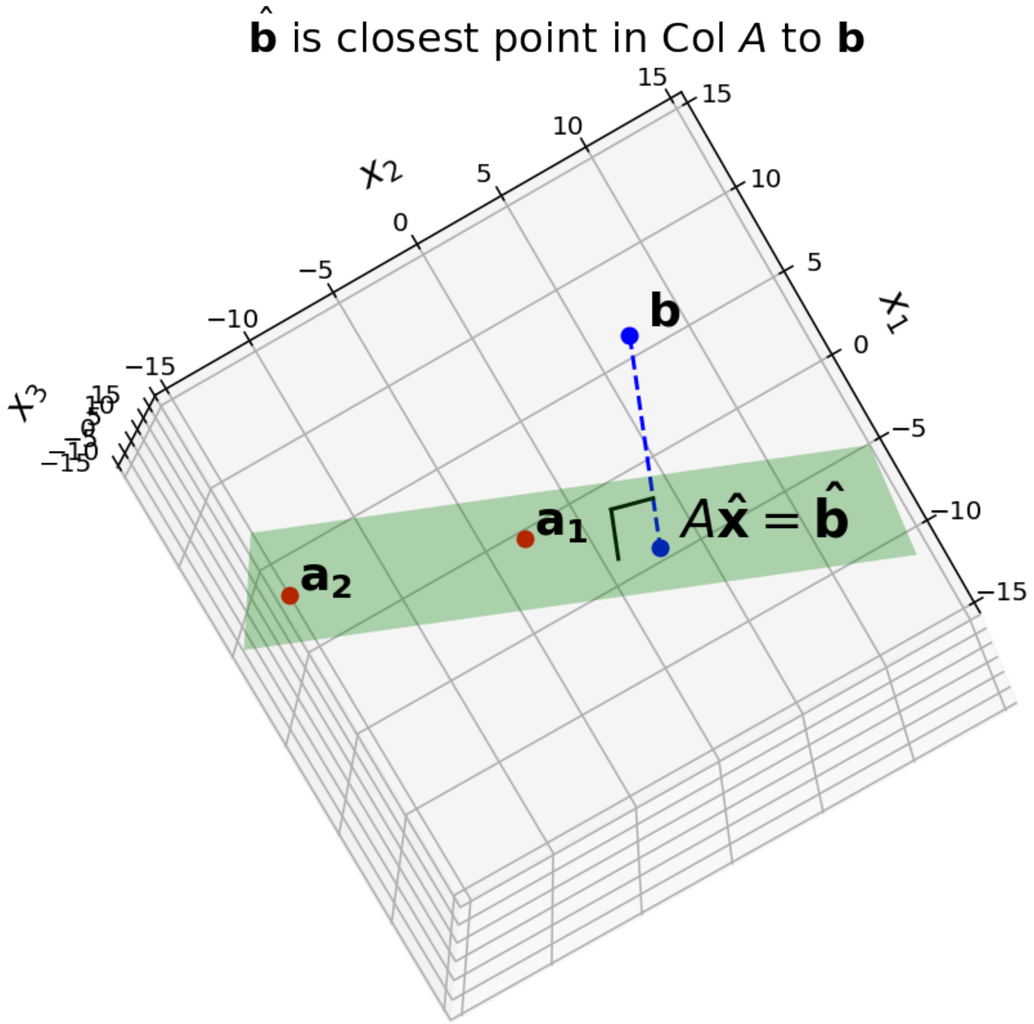
Q. Find α s.t. $\alpha\mathbf{u}$ is as close as possible to $\mathbf{y}$

i.e., $dist(\mathbf{y}, \alpha\mathbf{u}) = \|\mathbf{y} - \alpha\mathbf{u}\|$ is as small as possible

We need to generalize this

The General Least Squares Problem

Figure 22.8

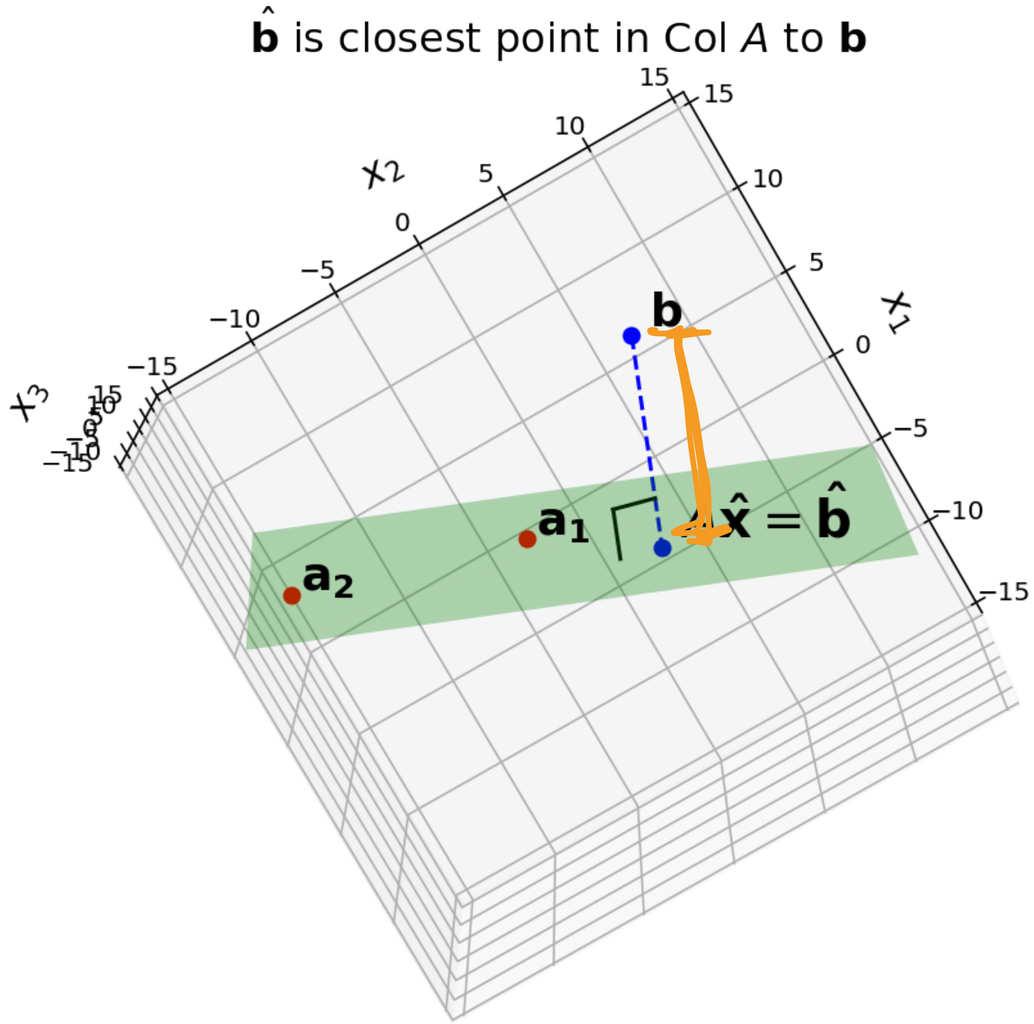


The General Least Squares Problem

Q. Given $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{b} \in \mathbb{R}^m$, find a vector $\mathbf{x} \in \mathbb{R}^n$ that *minimizes*

$$dist(A\mathbf{x}, \mathbf{b}) = \|A\mathbf{x} - \mathbf{b}\|$$

Figure 22.8



The General Least Squares Problem

Q. Given $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{b} \in \mathbb{R}^m$, find a vector $\mathbf{x} \in \mathbb{R}^n$ that *minimizes*

$$dist(A\mathbf{x}, \mathbf{b}) = \|A\mathbf{x} - \mathbf{b}\|$$

Find a vector $\mathbf{x}$ *which makes* $\|A\mathbf{x} - \mathbf{b}\|$ *as small as possible*

Figure 22.8

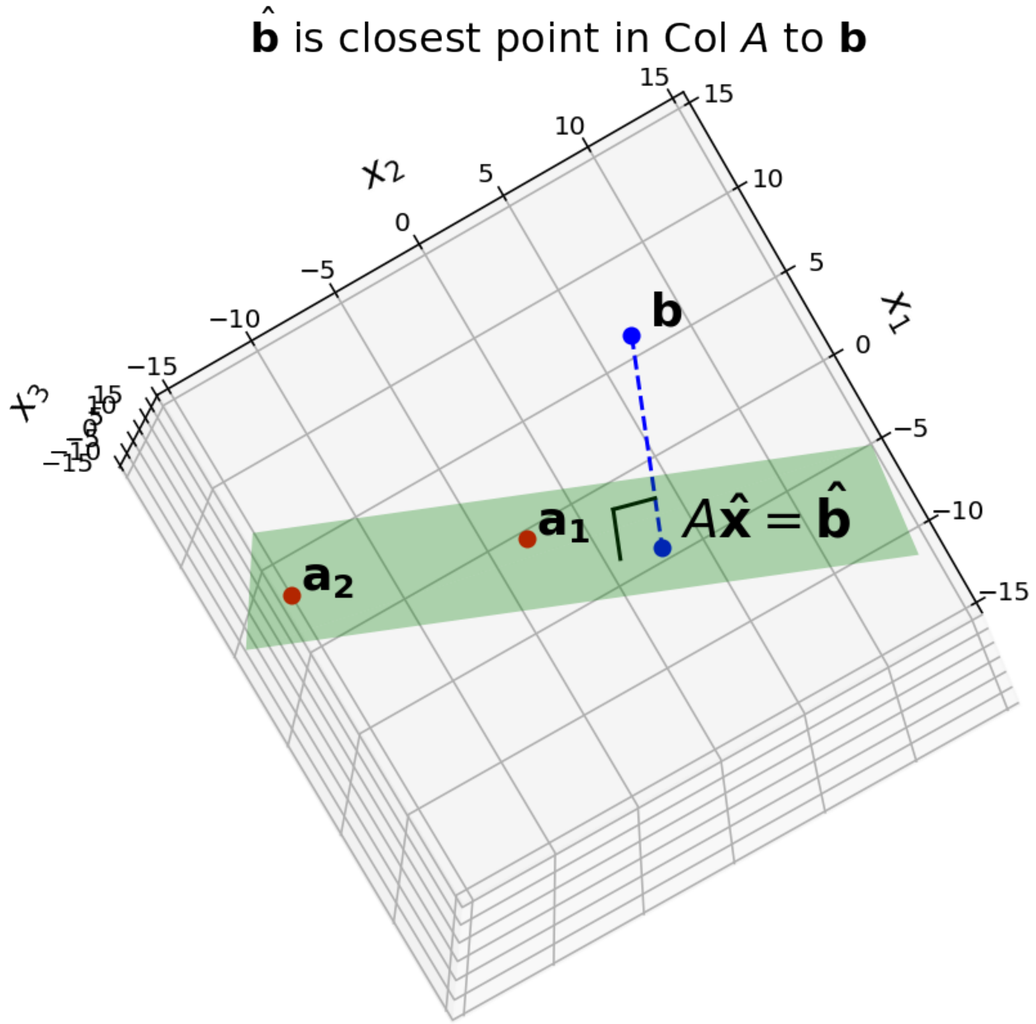
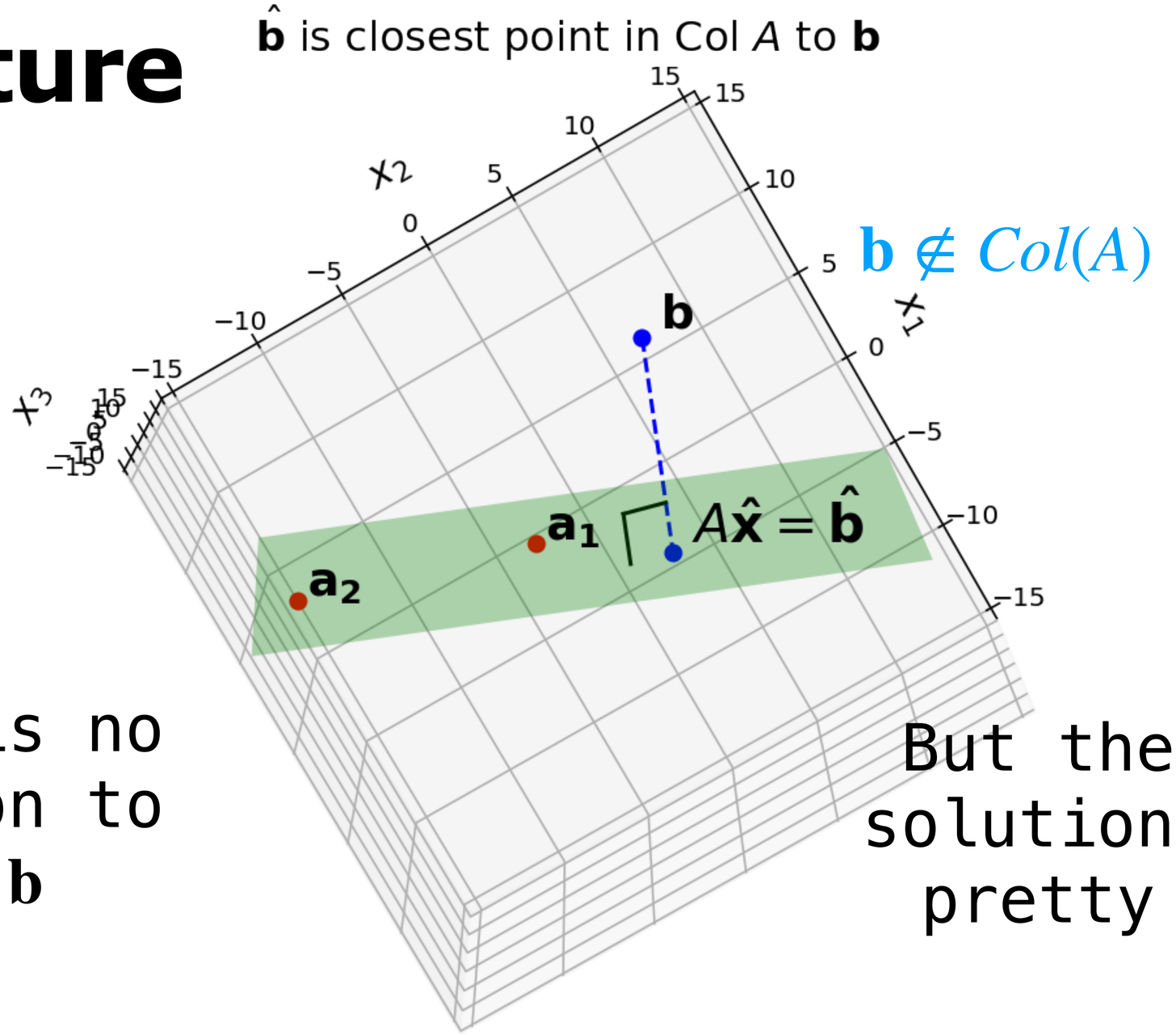


Figure 22.8

The Picture



There is no solution to $A\mathbf{x} = \mathbf{b}$

But there's a solution that's pretty close

Sum of Squares

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{n} ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

Sum of Squares

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{n} ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

It's equivalent to minimize $\|A\mathbf{x} - \mathbf{b}\|^2$

Sum of Squares

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{n} ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

It's equivalent to minimize $\|A\mathbf{x} - \mathbf{b}\|^2$

Sums of squares come up everywhere

Sum of Squares

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{n} ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

It's equivalent to minimize $\|A\mathbf{x} - \mathbf{b}\|^2$

Sums of squares come up everywhere

Note. This error is everywhere differentiable, whereas $\sum_{i=1}^{n} |(A\mathbf{x})_i - b_i|$ is not

Least Squares Solution

Def. Given $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{b} \in \mathbb{R}^m$, a **least squares solution** of $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ s.t.

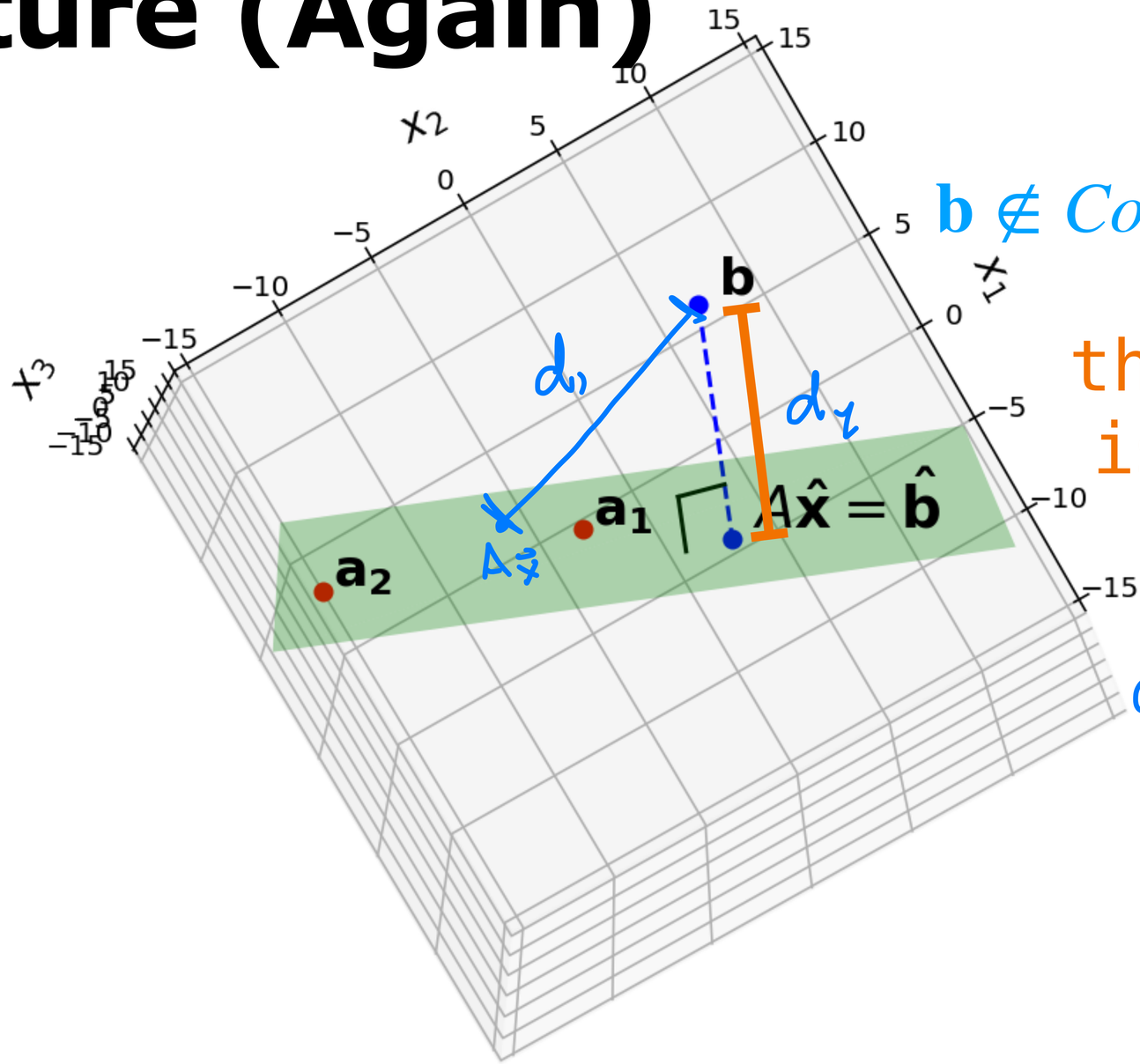
$$\|A\hat{\mathbf{x}} - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\|$$

for *any* $\mathbf{x}$ in $\mathbb{R}^n$

Again, $\|A\hat{\mathbf{x}} - \mathbf{b}\|$ is as small as possible

Figure 22.8

The Picture (Again)



Argmin

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|$$

Argmin

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|$$

Another way of framing this is via $\arg\min$

Argmin

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|$$

Another way of framing this is via $\arg\min$

Def. $\arg\min_{x \in X} f(x) = \hat{x}$ where $f(\hat{x}) = \min_{x \in X} f(x)$

Argmin

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|$$

Another way of framing this is via $\arg\min$

Def. $\arg\min_{x \in X} f(x) = \hat{x}$ where $f(\hat{x}) = \min_{x \in X} f(x)$

$\hat{x}$ is the ***arg****ument* that ***min****imizes* f

Argmin

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|$$

Another way of framing this is via $\arg\min$

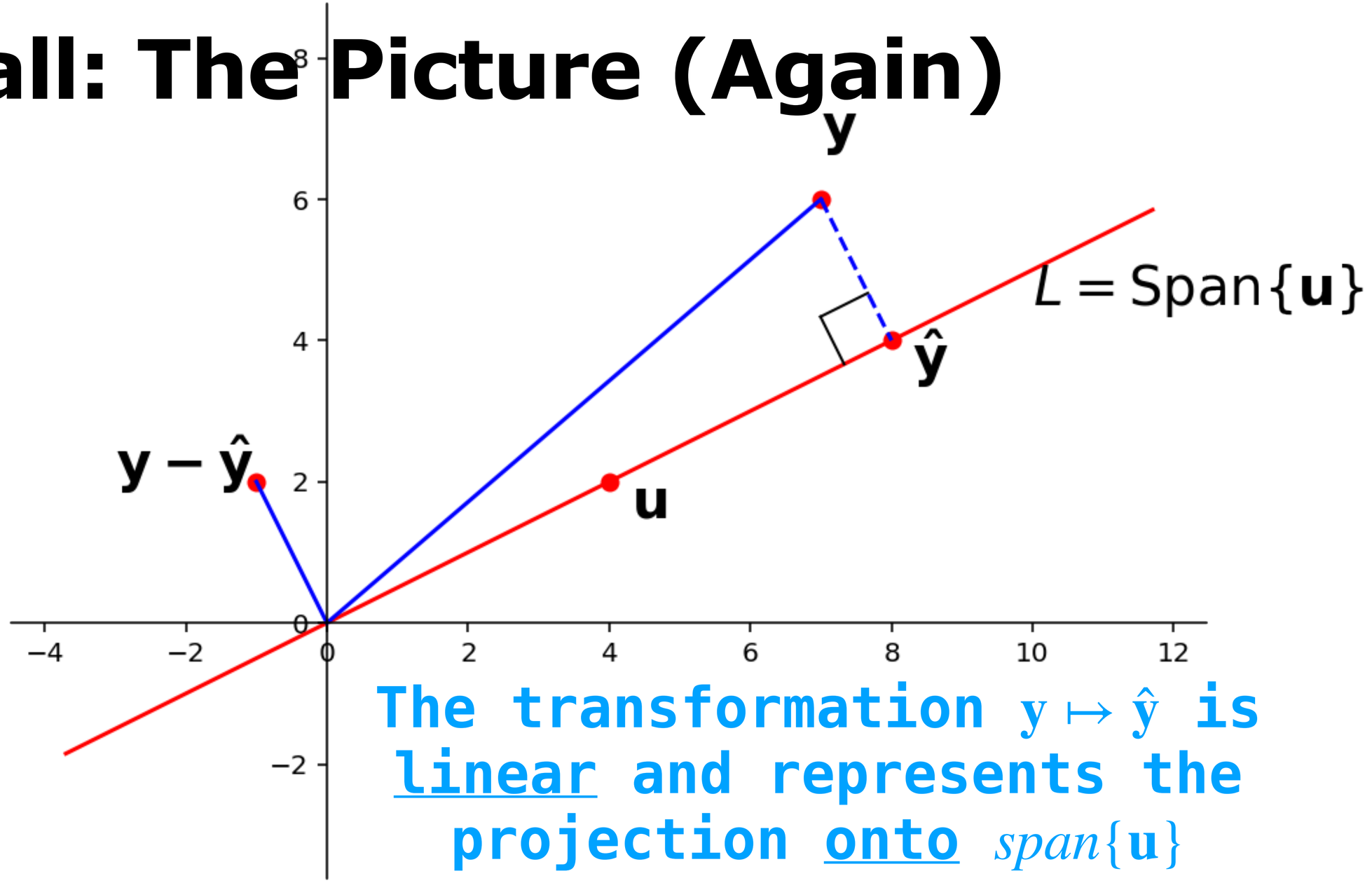
Def. $\arg\min_{x \in X} f(x) = \hat{x}$ where $f(\hat{x}) = \min_{x \in X} f(x)$

$\hat{x}$ is the ***arg****ument* that ***min****imizes* f

This is now an *optimization problem*

Solving the General Least Squares Problems

Recall: The Picture (Again)

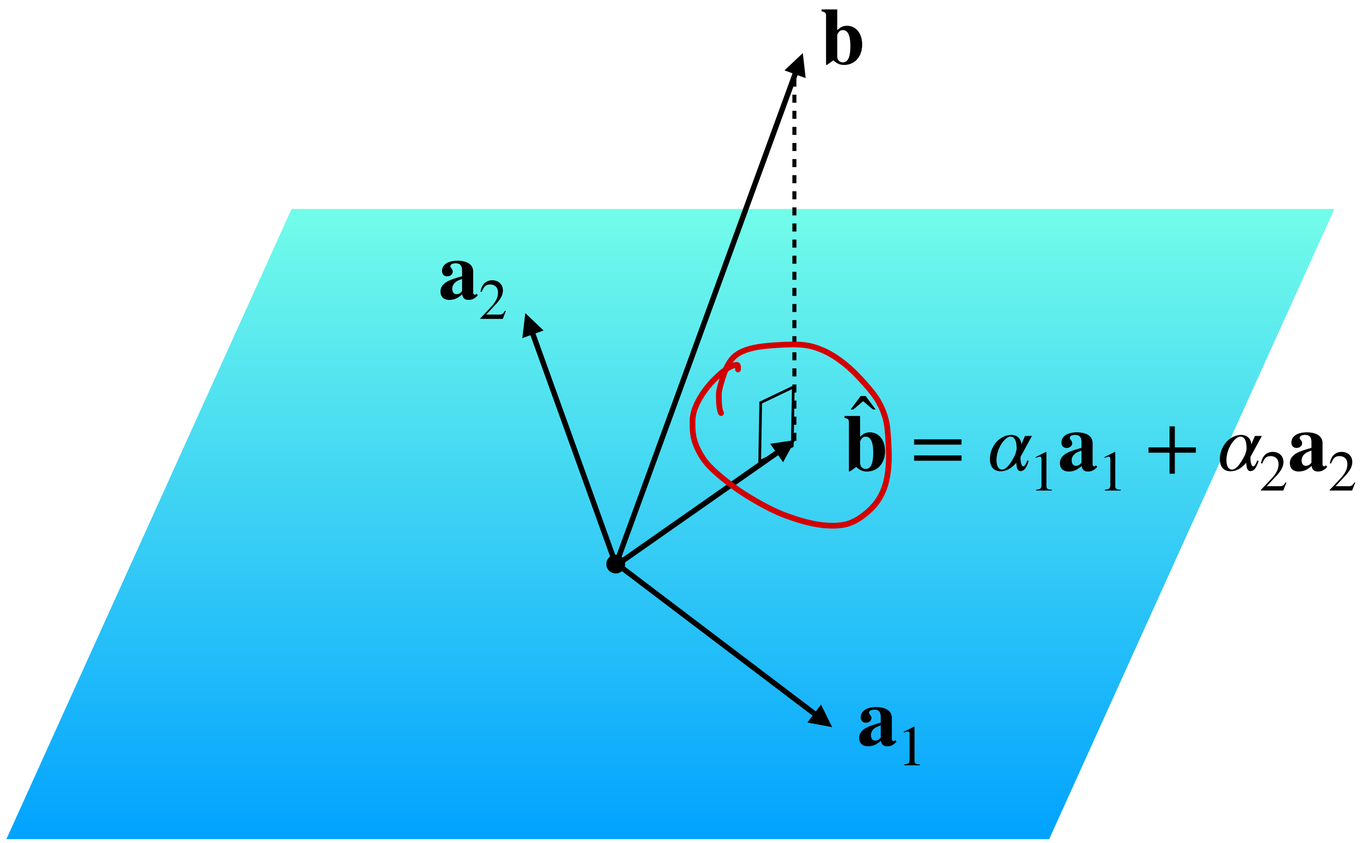


The transformation $\mathbf{y} \mapsto \hat{\mathbf{y}}$ is linear and represents the projection onto $span\{\mathbf{u}\}$

Projects onto other Spans

The transformation $\mathbf{b} \mapsto \hat{\mathbf{b}}$ is the projection of $\mathbf{b}$ onto $\text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$

basis



The High Level Approach.

Q. Find a least squares solutions to $A = \mathbf{b}$

A.

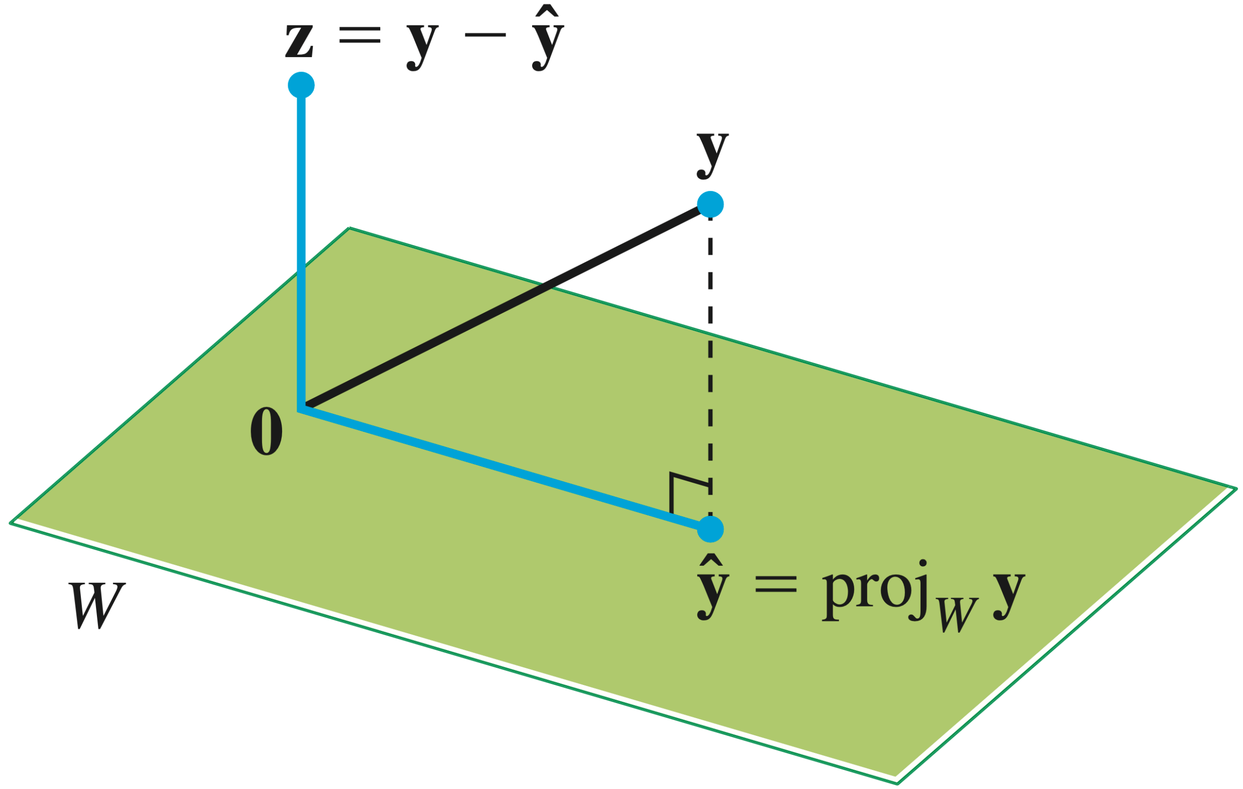
1. Find the closest point $\hat{\mathbf{b}}$ in $Col(A)$ to $\mathbf{b}$
2. Solve the equation $A\mathbf{x} = \hat{\mathbf{b}}$ instead

Orthogonal Decomposition Theorem

Thm. Let W be a subspace of $\mathbb{R}^n$. Every vector $\mathbf{y}$ in $\mathbb{R}^n$ can be written *uniquely* as

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

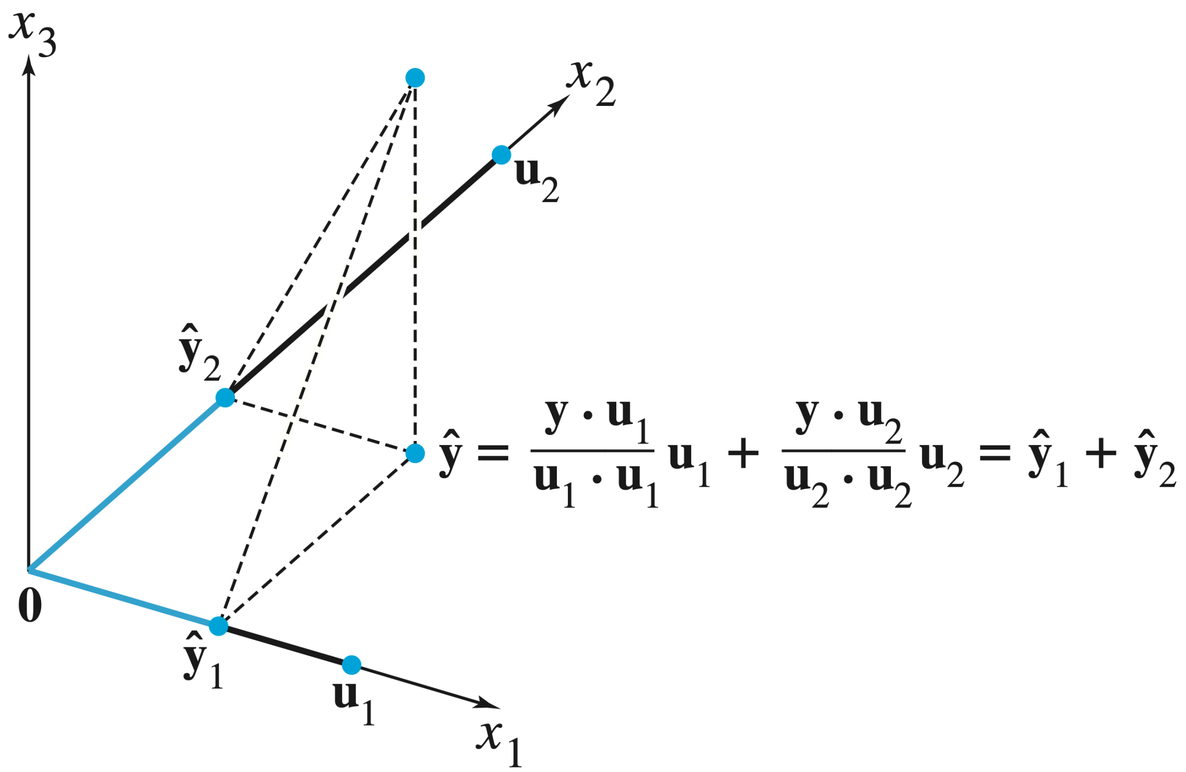
where $\hat{\mathbf{y}} \in W$ and $\mathbf{z}$ is orthogonal to every vector in W



Projection via Orthogonal Bases

We can determine $\hat{\mathbf{y}}$ by projecting onto an orthogonal basis

Every subspace has an orthogonal basis (we won't prove this)



Linear Algebra and its Applications, Lay, Lay, McDonald

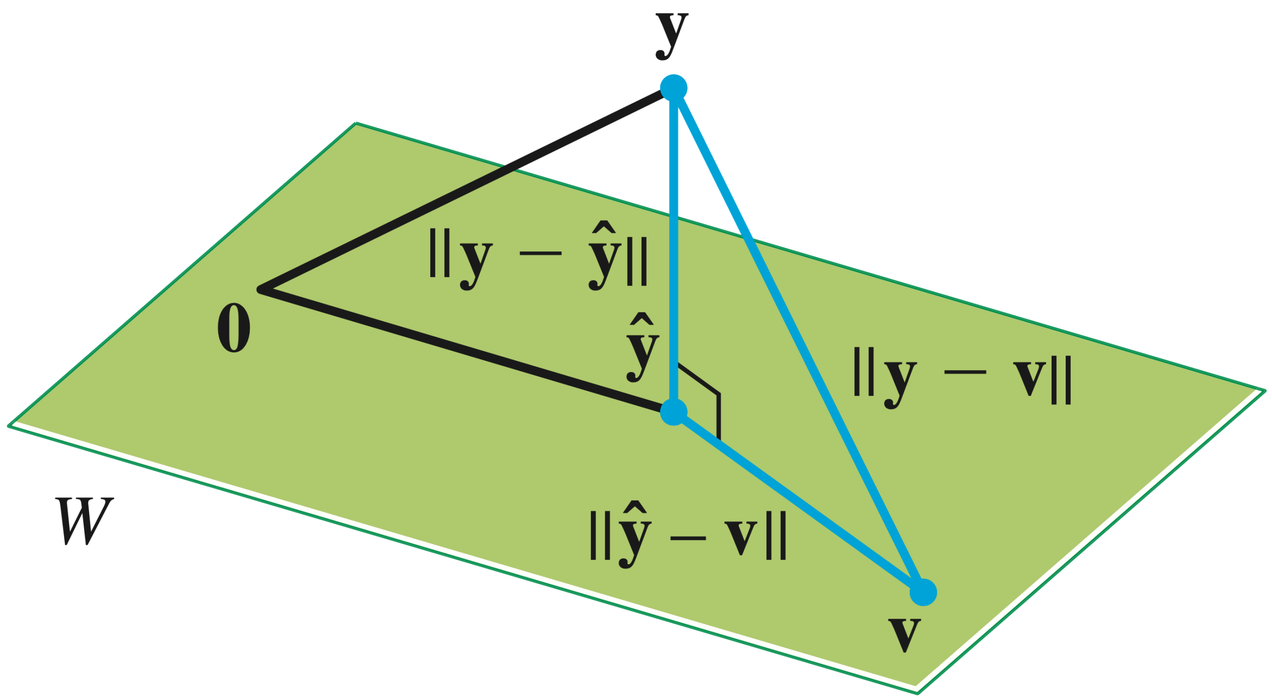
The Best-Approximation Theorem

Thm. Let W be a subspace of $\mathbb{R}^n$, and let $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W Then

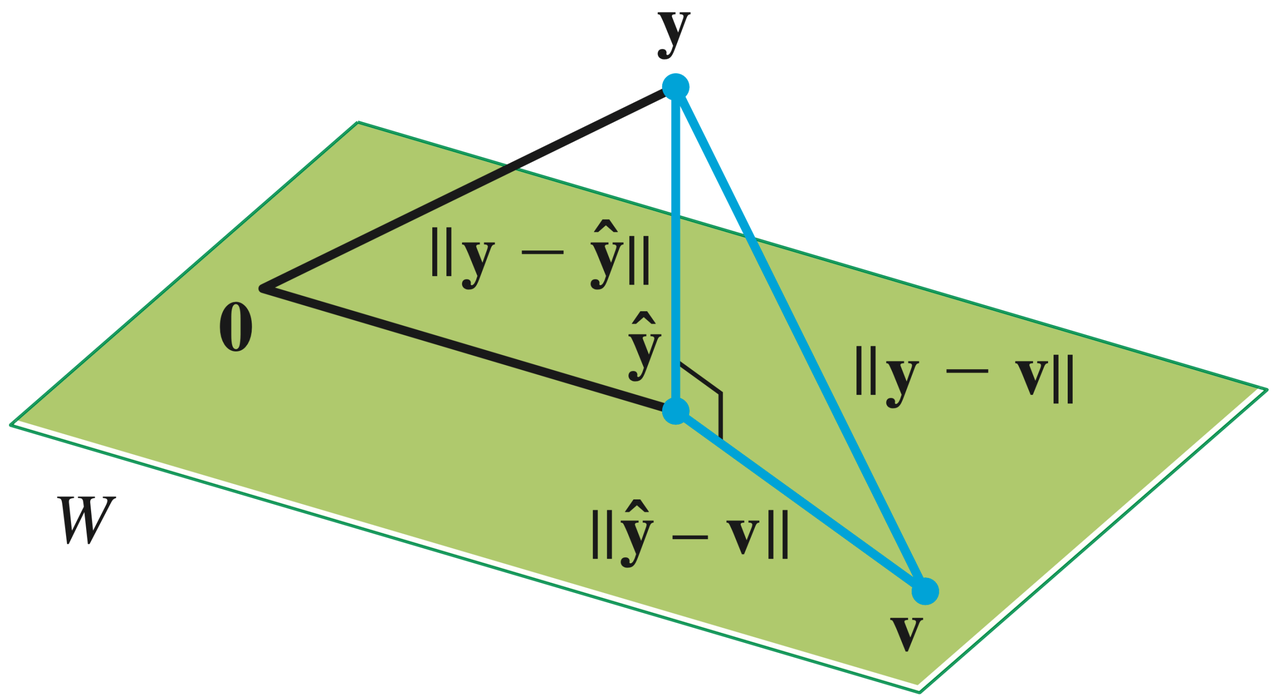
$$\|\mathbf{y} - \hat{\mathbf{y}}\| \leq \|\mathbf{y} - \mathbf{w}\|$$

for *any* vector $\mathbf{w}$ in W

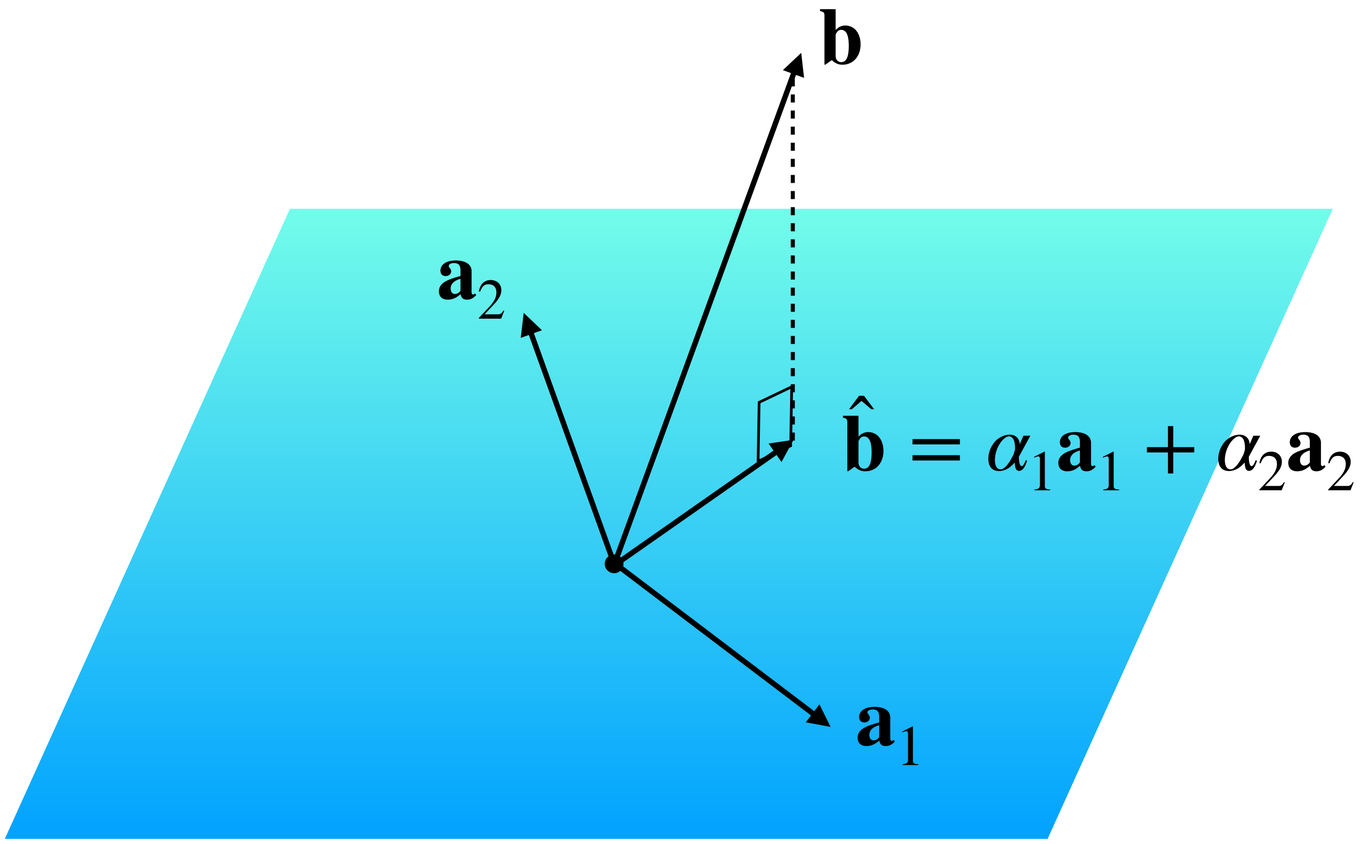
$\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$



Proof by Inspection

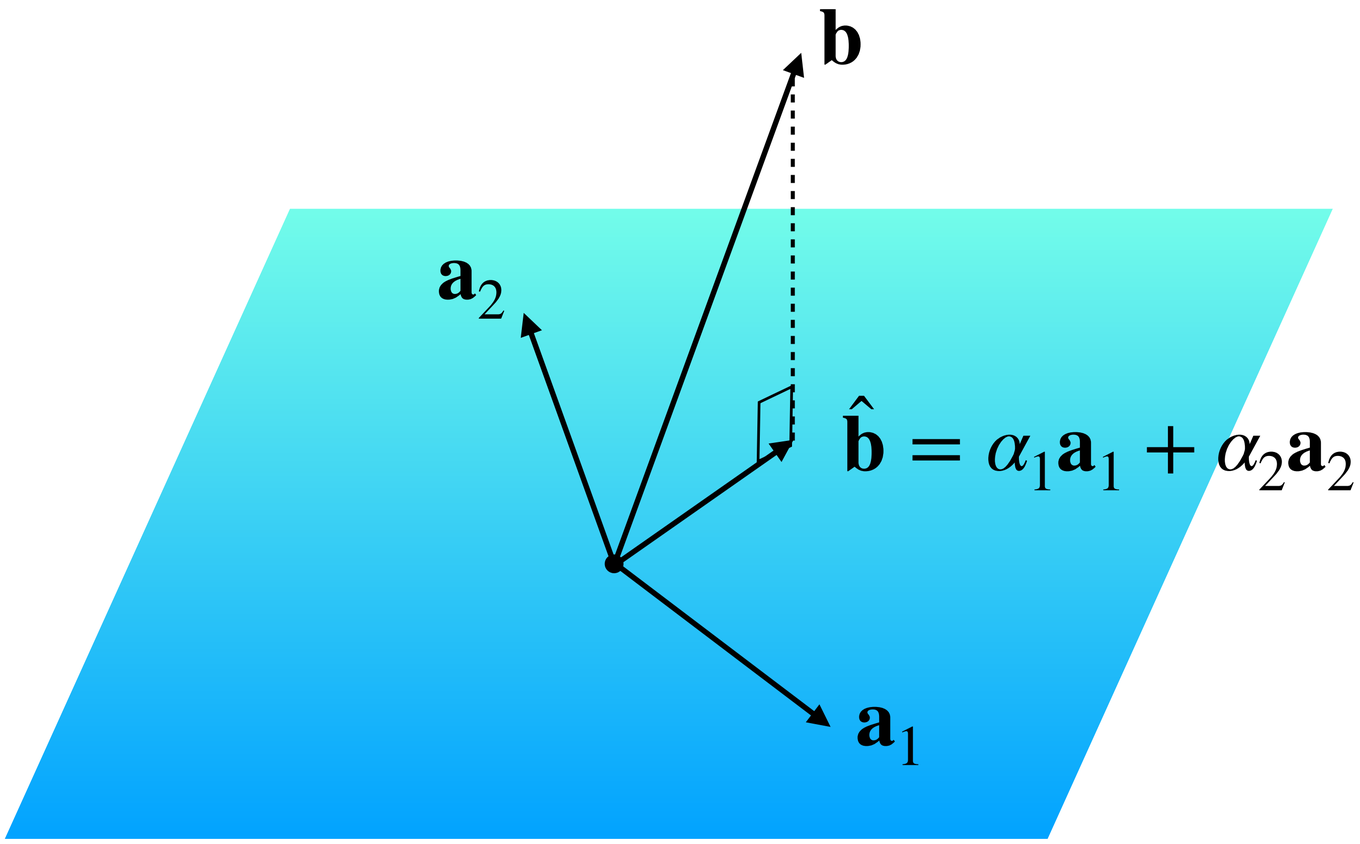


The Point


b
$\mathbf{a}_2$
$\hat{\mathbf{b}} = \alpha_1\mathbf{a}_1 + \alpha_2\mathbf{a}_2$
$\mathbf{a}_1$

The Point

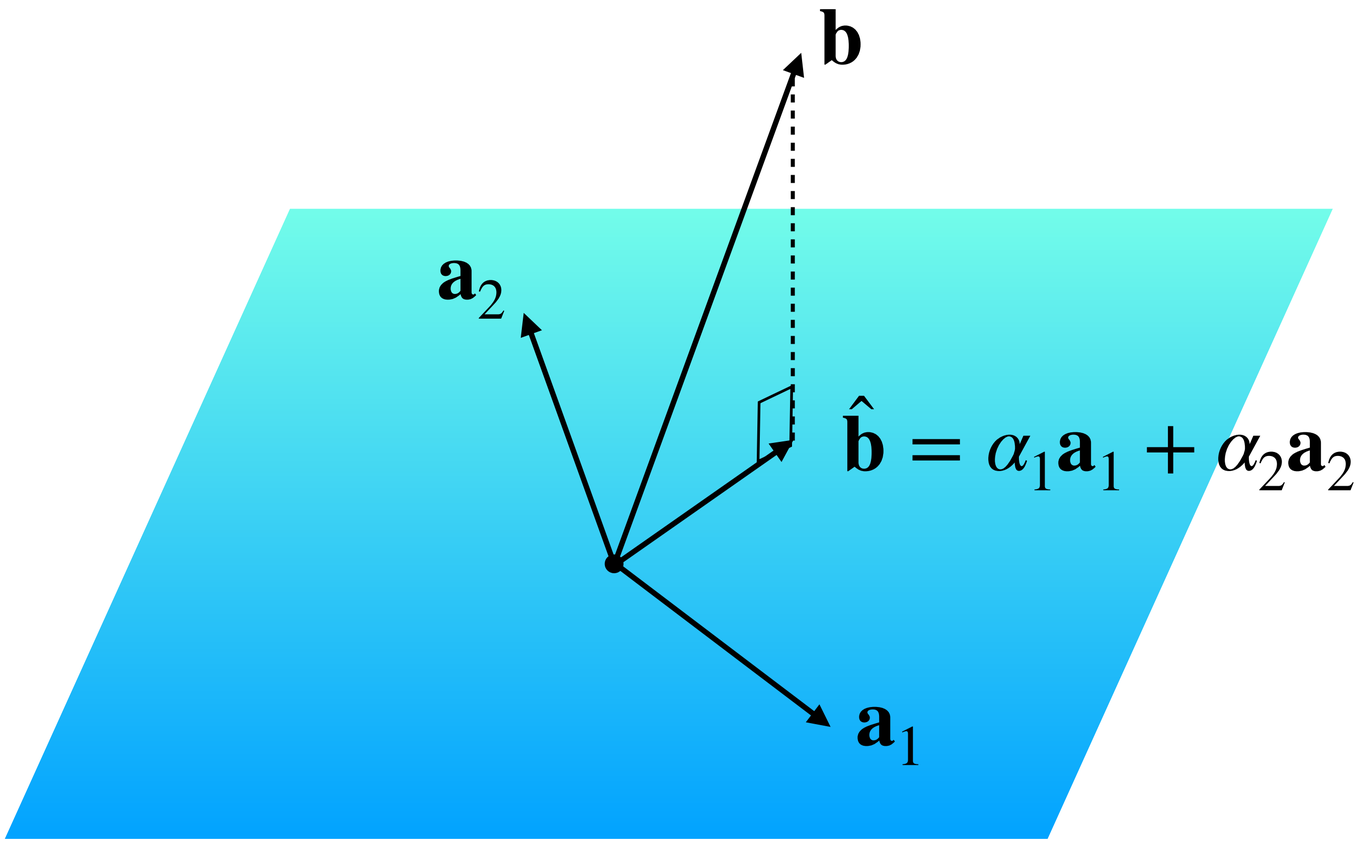
$\hat{\mathbf{b}}$ is in $Col(A)$ so $A\mathbf{x} = \hat{\mathbf{b}}$ has a solution



The Point

$\hat{\mathbf{b}}$ is in $Col(A)$ so $A\mathbf{x} = \hat{\mathbf{b}}$ **has a solution**

At this point, we could call it a day:

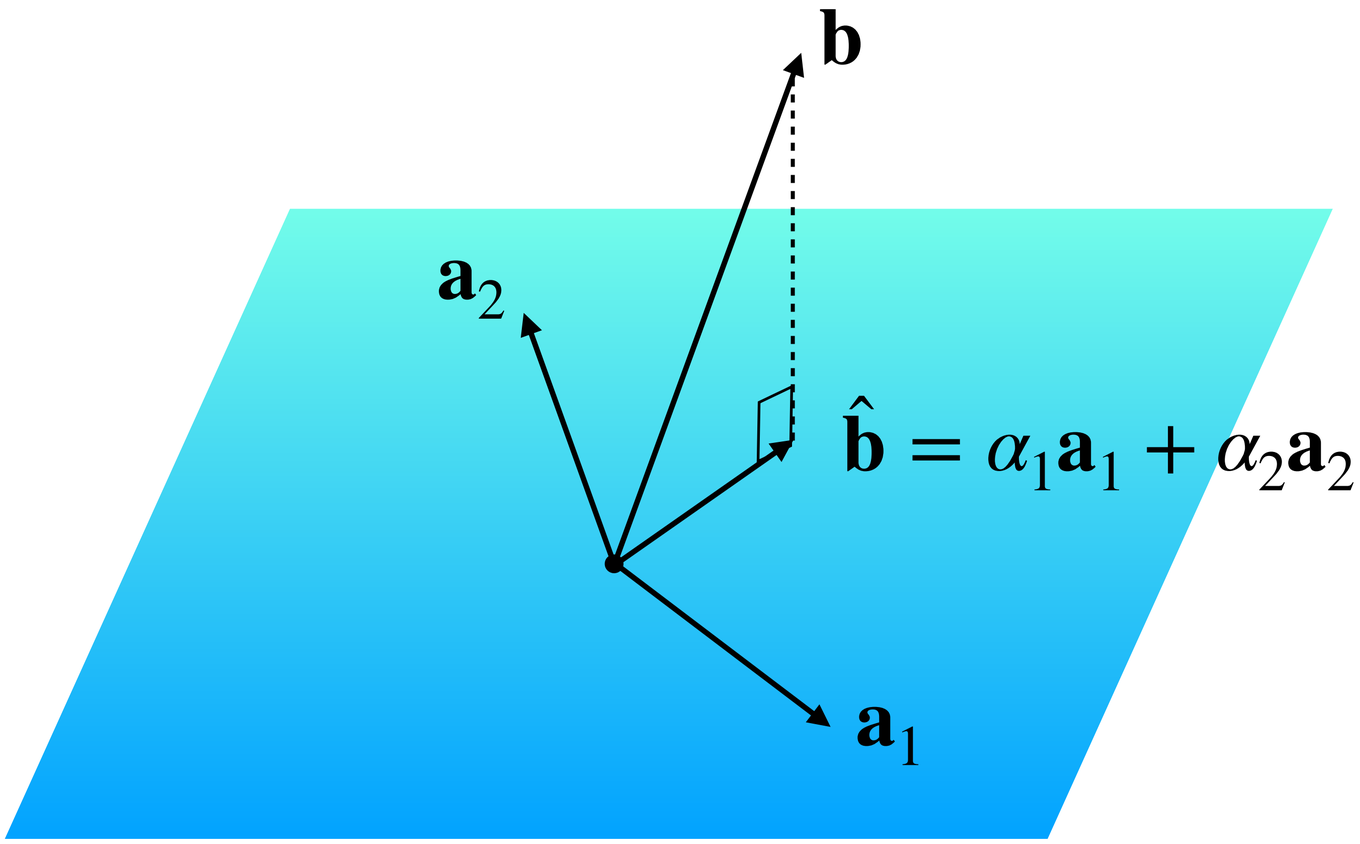


The Point

$\hat{\mathbf{b}}$ is in $Col(A)$ so $A\mathbf{x} = \hat{\mathbf{b}}$ **has a solution**

At this point, we could call it a day:

Q. Find a least squares solution to $A\mathbf{x} = \mathbf{b}$



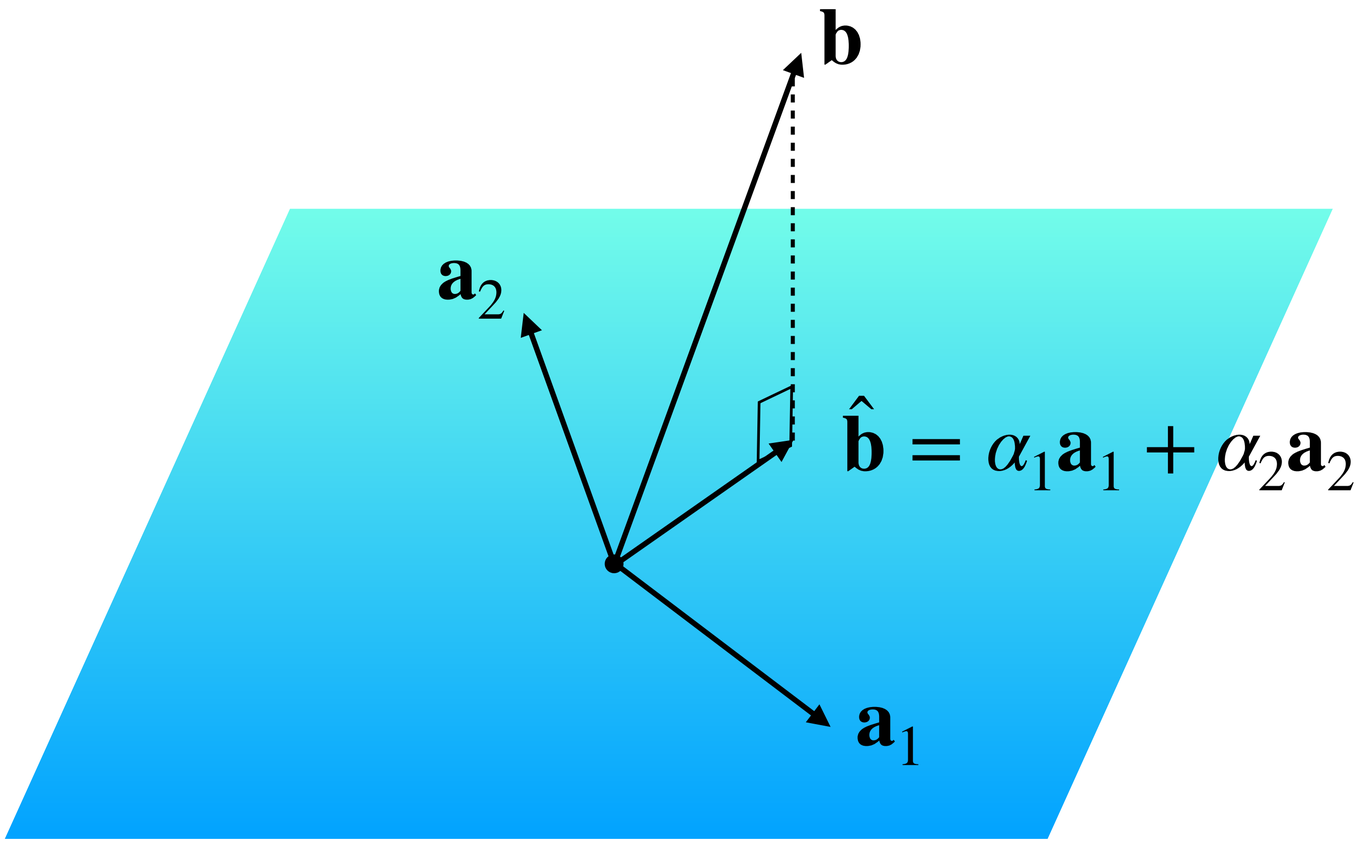
The Point

$\hat{\mathbf{b}}$ is in $Col(A)$ so $A\mathbf{x} = \hat{\mathbf{b}}$ **has a solution**

At this point, we could call it a day:

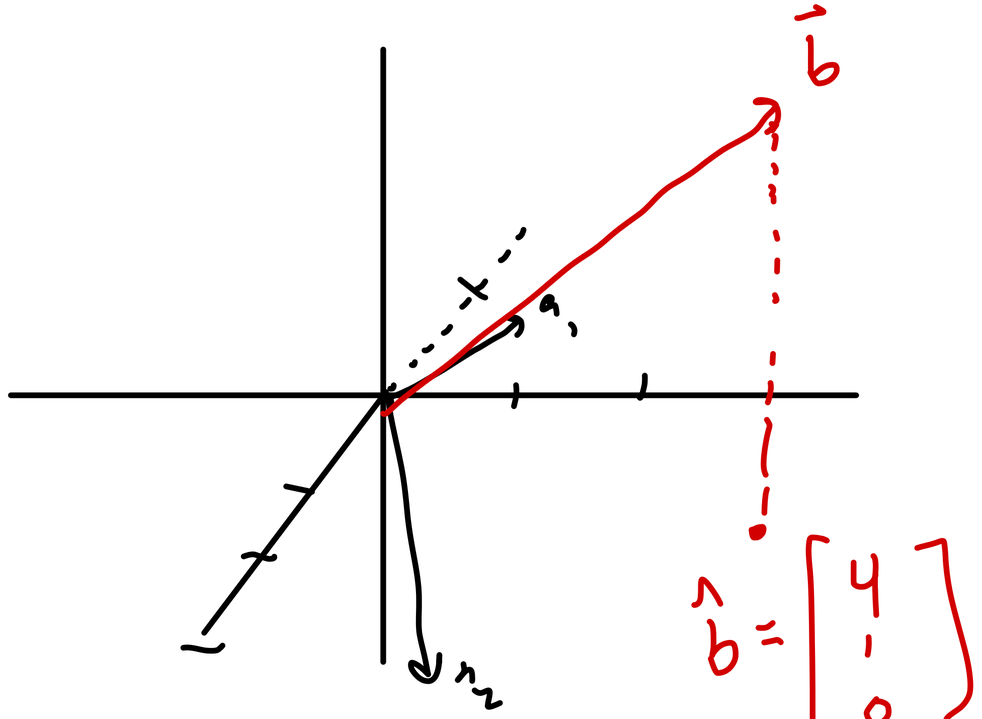
Q. Find a least squares solution to $A\mathbf{x} = \mathbf{b}$

A. Find $\hat{\mathbf{b}}$, then solve $A\mathbf{x} = \hat{\mathbf{b}}$



Example

$$\begin{bmatrix} a_1 & a_2 \\ 1 & 2 \\ -1 & 3 \\ 0 & 0 \end{bmatrix} \mathbf{x} = \begin{bmatrix} \vec{b} \\ 4 \\ 1 \\ 4 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 2 \\ -1 & -3 \\ 0 & 0 \end{bmatrix} \vec{x} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \hat{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

Exercise: solve

$$\hat{x} = \begin{bmatrix} y_1 \\ y_2 \\ 0 \end{bmatrix}$$

The Normal Equations

A Couple Observations

$$A = \begin{bmatrix} | & & | \\ \vec{a}_1 & \cdots & \vec{a}_n \\ | & & | \end{bmatrix}$$

$$A\hat{x} = \hat{b}$$

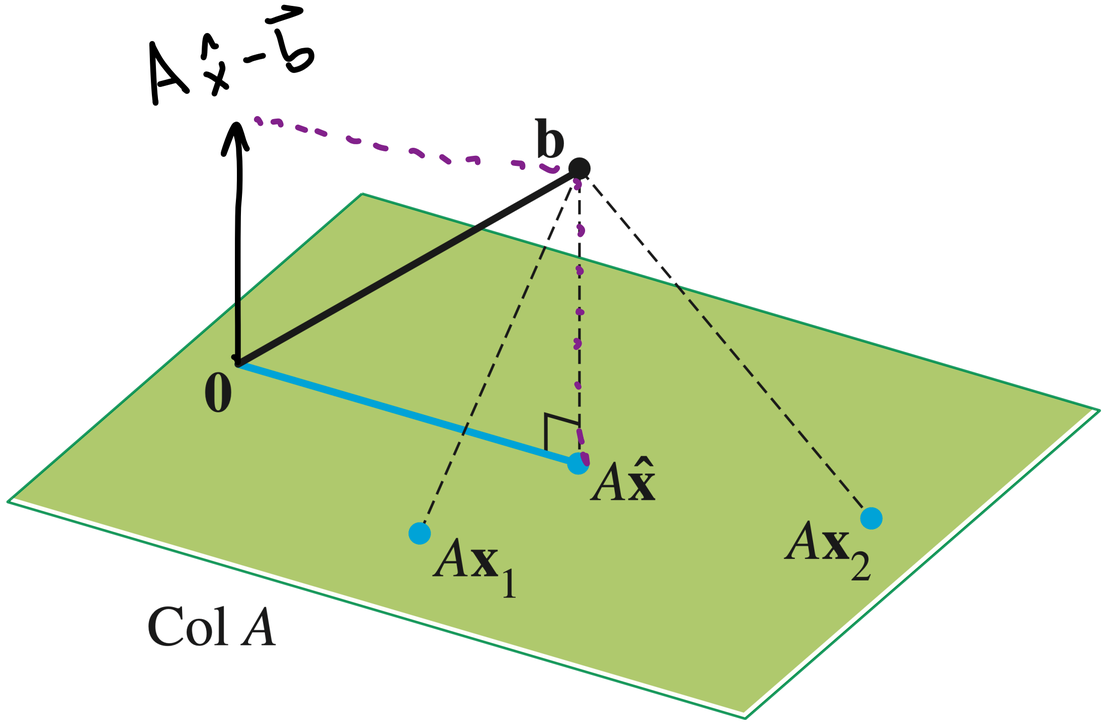
$$\hat{b} - b =$$

$$\langle A\hat{x} - b, \vec{r} \rangle = 0$$

$$\vec{r} \in \text{Col}(A)$$

$$\langle A\hat{x} - b, a_i \rangle = 0$$

$$\langle a_i, A\hat{x} - b \rangle = a_i^T (A\hat{x} - b) = 0$$



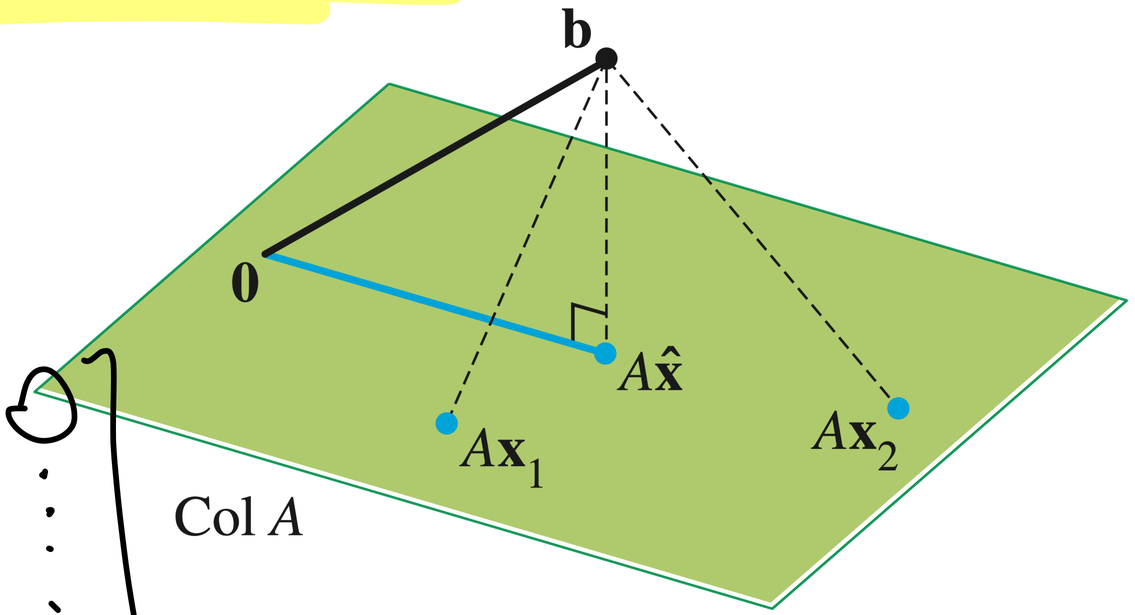
A Couple Observations

$$A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$$

$$A^T(A\hat{x} - b) = \vec{0}$$

$$a_1^T(A\hat{x} - b) = 0$$

$$A^T = \begin{bmatrix} — & a_1^T & — \\ & \vdots & \\ — & a_n^T & — \end{bmatrix} \quad (A\hat{x} - b) = \begin{bmatrix} | \\ | \\ | \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$



Simplify

$$A^T(A\hat{\mathbf{x}} - \mathbf{b})$$

$$A^T(A\hat{x} - b) = A^T A\hat{x} - A^T b = \vec{0}$$

$$A^T A\hat{x} = A^T b$$

Normal equation

The Normal Equations

The Normal Equations

Thm. The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ is the same as the set of solutions to

$$A^T A\mathbf{x} = A^T\mathbf{b}$$

The Normal Equations

Thm. The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ is the same as the set of solutions to

$$A^T A\mathbf{x} = A^T\mathbf{b}$$

In particular, this set of solutions is nonempty

The Normal Equations

Thm. The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ is the same as the set of solutions to

$$A^T A\mathbf{x} = A^T\mathbf{b}$$

In particular, this set of solutions is nonempty

(We just showed that if $\hat{\mathbf{x}}$ is a least squares solution then $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$)

Example

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$(A^T A)^T = A^T (A^T)^T = A^T A$

$2 \times 3 \quad 3 \times 2 \quad 2 \times 2$

$$A^T b = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \hat{x} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 0 & 0 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 4 \end{bmatrix}$$

Unique Least Squares Solutions

Question

Is a least squares solution unique?

Answer: No

If $\mathbf{b} \in Col(A)$ then $\hat{\mathbf{b}} = \mathbf{b}$ and then we're asking if $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of A

$$A^T Ax = A^T b$$

When is there a unique solution?

The least squares method gives us to find an approximate solution when there is no exact solution

But it doesn't help us choose a solution in the case that there are many

Practically Speaking

numpy.linalg.lstsq

linalg.lstsq(*a*, *b*, *rcond*='*warn*') [source]

Return the least-squares solution to a linear matrix equation.

Computes the vector *x* that approximately solves the equation `a @ x = b`. The equation may be under-, well-, or over-determined (i.e., the number of linearly independent rows of *a* can be less than, equal to, or greater than its number of linearly independent columns). If *a* is square and of full rank, then *x* (but for round-off error) is the "exact" solution of the equation. Else, *x* minimizes the Euclidean 2-norm $||b - ax||$. If there are multiple minimizing solutions, the one with the smallest 2-norm $||x||$ is returned.

Parameters: **a** : *(M, N) array_like*

"Coefficient" matrix.

b : *{(M,), (M, K)} array_like*

Ordinate or "dependent variable" values. If *b* is two-dimensional, the least-squares solution is calculated for each of the *K* columns of *b*.

rcond : *float, optional*

Practically Speaking

numpy.linalg.lstsq

`linalg.lstsq(a, b, rcond='warn')` [source]

Return the least-squares solution to a linear matrix equation.

Computes the vector *x* that approximately solves the equation `a @ x = b`. The equation may be under-, well-, or over-determined (i.e., the number of linearly independent rows of *a* can be less than, equal to, or greater than its number of linearly independent columns). If *a* is square and of full rank, then *x* (but for round-off error) is the "exact" solution of the equation. Else, *x* minimizes the Euclidean 2-norm $||b - ax||$. If there are multiple minimizing solutions, the one with the smallest 2-norm $||x||$ is returned.

NumPy chooses the shortest vector

Parameters: **a** : *(M, N) array_like*

"Coefficient" matrix.

b : *{(M,), (M, K)} array_like*

Ordinate or "dependent variable" values. If *b* is two-dimensional, the least-squares solution is calculated for each of the *K* columns of *b*.

rcond : *float, optional*

Practically Speaking

numpy.linalg.lstsq

`linalg.lstsq(a, b, rcond='warn')` [source]

Return the least-squares solution to a linear matrix equation.

Computes the vector *x* that approximately solves the equation `a @ x = b`. The equation may be under-, well-, or over-determined (i.e., the number of linearly independent rows of *a* can be less than, equal to, or greater than its number of linearly independent columns). If *a* is square and of full rank, then *x* (but for round-off error) is the "exact" solution of the equation. Else, *x* minimizes the Euclidean 2-norm $||b - ax||$. If there are multiple minimizing solutions, the one with the smallest 2-norm $||x||$ is returned.

NumPy chooses the shortest vector

(why?...)

Parameters: **a** : *(M, N) array_like*

"Coefficient" matrix.

b : *{(M,), (M, K)} array_like*

Ordinate or "dependent variable" values. If *b* is two-dimensional, the least-squares solution is calculated for each of the *K* columns of *b*.

rcond : *float, optional*

Unique Least Squares Solutions

Thm. For $A \in \mathbb{R}^{m \times n}$ the following are equivalent:

- $A\mathbf{x} = \mathbf{b}$ has a *unique* least squares solution for any choice of $\mathbf{b}$
- The columns of A are *linearly independent*
- $A^T A$ is *invertible*

Unique Least Squares Solutions

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

If A has linearly independent columns, then its unique least squares solution is the above

$$A^T A \hat{x} = A^T \vec{b}$$

Projecting onto a subspace

$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = A(A^TA)^{-1}A^T\mathbf{b}$$

If the columns of A are linearly independent, then **they form a basis**

Said another way: if $\mathscr{B}$ is a basis, then we can construct a matrix A whose columns are the vectors in $\mathscr{B}$

This means we can find arbitrary projections